Questions Before Priors:
Causal Dissonance, Insight, and the Origin of Strategic Theories*

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— First draft —

Abstract

The theory-based view treats economic actors as theorists whose causal understandings direct search, experimentation, and action. Existing formal treatments begin after the theory exists. This paper models where strategic theories come from. Agents represent their environment through awareness partitions of a state space; expressible causal theories assign elementary mechanisms to awareness cells; theories and beliefs determine committed policies; and the policy profile causes everyone's consequences through an objective causal system. Bayesian learning operates only within the current language, and a misspecified posterior grows confident rather than alarmed, so detecting representational failure requires a distinct faculty: causal dissonance, the statistical rejection of every expressible theory. Dissonance directs inquiry without determining its output; a costly, fallible technology may produce a previously inexpressible distinction; and a bridge prior restores Bayesian competence on the expanded language. An agent can estimate the value of a question before its answers are conceivable, warrant is distinct from post-insight confidence—the insight itself is not evidence—and an agent generally cannot measure its own overconfidence. Because theories cause actions and actions generate evidence, inquiry dies out almost surely, leaving every agent either in a Bayesian world or in rationally tolerated dissonance—its question unpriced, unanswerable, or abandoned. The configurations that remain can preserve false theories, heterogeneous awareness, and permanent performance differences: shared false theories survive exactly when neither strategic variety nor environmental harshness exposes them, and performance gaps persist whenever the laggard's question is priced below the cost of asking.

Keywords: theory-based view; unawareness; insight; causal learning; subjective rationality; self-confirming equilibrium

*Preliminary draft; comments welcome. This paper develops the formal core of a research program on pre-Bayesian theory formation in strategy.

1 Introduction

The theory-based view of strategy holds that economic actors are theorists: managers and entrepreneurs carry causal accounts of how value is created, and those accounts—not merely their information or their incentives—direct what they notice, which experiments they run, and what they do (Felin and Zenger, 2017; Felin et al., 2024). A decade of work has established the consequences of holding a theory. Theories guide experimentation and search (Camuffo et al., 2020; Chavda et al., 2024), support the acquisition of resources and rents (Ehrig and Zenger, 2024), and discipline entrepreneurial decision-making in the field (Camuffo et al., 2024). What this literature has not yet formalized is the step before all of that: where a genuinely new theory comes from. Existing formal treatments begin with the theory present—as a menu of candidate theories to select among (Camuffo et al., 2024), as a given causal conjecture that directs search (Chavda et al., 2024), or as a given awareness advantage to be exploited competitively (Ehrig and Zenger, 2024). The origin of the theory is described verbally and documented empirically (Ott and Hannah, 2024), and the absence of formal machinery for it has been explicitly noted as the field's open analytical problem (Felin et al., 2024).

The difficulty is not a missing application of Bayesian reasoning; it is a boundary of Bayesian reasoning. A Bayesian agent updates beliefs over a space of represented possibilities. Conditioning reweights that space and cannot enlarge it, so no sequence of updates makes an unrepresented causal distinction available for belief or action. Worse, when every represented theory is false, the posterior does not sound an alarm: by the logic of misspecified Bayesian learning (Berk, 1966), it concentrates on the least wrong of the available theories, and the agent grows more confident, not less. Within-language updating is self-sealing. The economics of growing awareness has developed axiomatic accounts of how beliefs should extend when the state space expands (Karni and Vierø, 2013, 2017; Vierø, 2021), but in those models the expansion itself arrives exogenously; models of discovery in games describe how play can reveal objectively available actions (Schipper, 2021) without explaining how experience generates a substantive causal question or its answer. What is missing, in both the strategy literature and the economics of unawareness, is a model of the transition: from a broken causal language, through a directed question, to a new representation, and back to ordinary Bayesian life.

This paper models that transition, embedded in strategic interaction. Its organizing object is the chain

$$\begin{aligned} &\text{theory} \longrightarrow \text{Bayesian judgment} \longrightarrow \text{action} \longrightarrow \text{experience} \longrightarrow \text{dissonance} \longrightarrow \\ &\quad \text{inquiry} \longrightarrow \text{refined awareness} \longrightarrow \text{new models} \longrightarrow \text{theory} \longrightarrow \cdots \end{aligned}$$

the process by which an agent whose causal representation has failed regains the capacity to be a Bayesian. The back end of the model follows the subjective-rationality tradition (Kalai and Lehrer, 1993b; Ryall, 2003): a finite set of agents adopt causal theories, combine them with beliefs and conjectures to choose committed policies, and the policy profile determines every agent's consequences through an objective causal system that absorbs all market and environmental detail. The front end is new. Agents represent the environment through awareness partitions of a state space; the theories they can formulate assign elementary causal mechanisms to awareness cells; and the partition itself—the language—is endogenous. An awareness cell is an implicit claim that the states it pools are causally alike. Strategic experience can refute that claim: when the statistical record rejects every expressible theory, the agent faces *causal dissonance*, a determinate defect with no expressible repair. Dissonance makes a question salient without providing its answer. A costly, fallible, truth-blind inquiry technology may then produce a refinement of the partition—the formal footprint of an insight—after which a bridge prior restores a probability space and Bayesian judgment resumes on the expanded language.

Five sets of results give the chain its content. First, boundary results locate exactly where Bayesian competence stops: conditioning cannot expand the language, and the misspecified posterior is self-sealing, so detecting representational failure requires a faculty formally distinct from updating. Second, dissonance directs inquiry without determining it. The record restricts admissible repairs—every restorative refinement must separate the evidence clusters that no single mechanism can accommodate—but generally many minimal repairs remain, observationally equivalent worlds demand different answers, and no truth-blind technology can be uniformly correct. Fallibility is the price of a representation that economizes on distinctions. Third, an agent can price a question before its answers are conceivable. A leverage statistic built from the agent's own retained experience—its remembered tokens and the coherent groupings it can articulate, never the analyst's list of repairs—estimates the value of inquiry at exactly the moment the posterior is undefined, and it tracks whether an answer could change action, not how much uncertainty remains. The estimate is honest: under noise the agent cannot tell a causal contrast from sampling accident, and the achievability weight it applies decays as inquiry fails. Questions die rationally: sustained success extinguishes them, defeat concentrates the evidence and the incentive in the loser, and hard questions are eventually abandoned. Fourth, the return to Bayes is disciplined by a warrant account: because any successful repair fits the experience that generated it by construction, retrospective fit carries no information, and—when inquiry is truth-blind—neither does the arrival of the insight itself. Post-insight confidence therefore exceeds warrant by a gap of size an outside assessor can compute, and an agent holding a minimal repair generally cannot even formulate the comparison that would measure its own gap. Fifth, because theories cause actions and actions generate the evidence, the population dynamics close the loop. Almost surely, awareness change and inquiry die out, leaving every agent either inhabiting a Bayesian world—models fit, questions dead—or in rationally tolerated dissonance, its question unpriced, unanswerable, or abandoned. We call the fully-Bayesian configurations *self-confirming unawareness equilibria*, and their interest is epistemic rather than strategic: such a configuration can preserve false theories and heterogeneous awareness, and a shared false theory survives precisely when neither of the two corrective forces—strategic variety inside an agent's categories, or an environment harsh enough to punish misservice observably—is present.

The persistence results deserve emphasis because of what carries them. Performance differences persist in this model not because an equilibrium concept freezes them but because the process that would erase them has died economically: the disadvantaged agent's question generates no leverage, loses its achievability weight, or meets an irreparable record—in each case it is priced below the cost of asking. Strategic variety is therefore traced to its source. Agents with identical preferences, identical data-processing competence, and identical objective opportunities end up with different causal languages, different theories, and different payoffs because their experiences posed different questions, their technologies produced different repairs, and their subsequent positions extinguished further inquiry at different points. This is a microfoundation for heterogeneity that begins before probability does.

A deliberately small example runs through the paper—two firms, four customer states, two product architectures—in which every object can be computed exactly: the four minimal repairs of a broken language, the two that survive a simplicity filter, the impossibility of choosing between them from the data, the exact price of the question, the exact warrant gap, and the knife-edge condition under which a market's shared false theory protects itself forever. The example is a laboratory, not the theory; each of its results instantiates a general statement.

The paper proceeds as follows. Section 2 locates the contribution. Section 3 presents the model. Section 4 develops the Bayesian layer and its boundary; section 5 defines dissonance and exposure; section 6 analyzes restoration, direction, and fallibility; section 7 prices inquiry; section 8 develops formulation, warrant, and the return to Bayesian competence; and section 9 closes the loop with the population dynamics. Section 10 discusses implications and the research program.

2 Related literature

The theory-based view. The foundational claim is that theories do more than summarize evidence: they direct attention toward causal possibilities, support counterfactual reasoning, and guide experimentation (Felin and Zenger, 2017). The *Strategy Science* collection on theory-based decisions substantially developed the view (Felin et al., 2024): Camuffo et al. (2024) formalize the choice of which theory to test and show that the value of experimentation grows with the number and uncertainty of candidate theories; Chavda et al. (2024) model search guided by a given causal theory; Ehrig and Zenger (2024) translate theories into awareness structures and show how partial disclosure secures resources and rents; Sorenson (2024) and Adner and Levinthal (2024) probe the mechanisms and the limits of theory-guided experimentation; and Ott and Hannah (2024) document empirically how entrepreneurs develop a poorly formed thesis into an articulated causal model. A parallel empirical program shows that training entrepreneurs to formulate and test theories improves performance (Camuffo et al., 2020). Related conceptual work presses the same open question this paper addresses: Zellweger and Zenger (2023) treat entrepreneurs as scientists generating beliefs under uncertainty, and Felin and Holweg (2024) argue that forward-looking causal theorizing is precisely what data-driven prediction cannot supply. Throughout, the formal treatments condition on the theory, the attribute list, or the awareness advantage being present. This paper supplies the front end those models presuppose, and its equilibrium analysis shows when the supplied front end matters: heterogeneous causal languages, and the performance differences they support, persist exactly when the strategic and environmental conditions identified below fail to erase them.

Unawareness and growing awareness. Standard state-space models cannot represent nontrivial unawareness (Dekel et al., 1998); generalized structures represent multiple awareness levels and interactive beliefs (Heifetz et al., 2006, 2013). On belief revision under expansion, reverse Bayesianism preserves relative odds as the state space grows (Karni and Vierø, 2013), with extensions to awareness of unawareness (Karni and Vierø, 2017), intertemporal settings (Vierø, 2021), unforeseen evidence (Piermont, 2021), and learning under incomplete awareness (Grant et al., 2022). Schipper (2021) distinguishes learning, which discards conceived possibilities, from discovery, which adds unconceived ones, and shows how play can move players to richer games; Schipper (2025) shows that beliefs about novelty can be held without representing the novel objects—a precedent for the delabeled constructs used here. Relative to this literature the paper endogenizes what those models take as given: the expansion event itself. Experience generates the defect, the defect localizes the question, a fallible technology produces the new distinction, and the bridge prior problem—which reverse Bayesianism constrains but does not solve—is embedded in an explicit account of what the generating evidence can and cannot warrant. In strategy, Bryan et al. (2022) bring unawareness into value capture and show that awareness itself earns rents; the present paper models where the awareness advantage originates, which that paper leaves open.

Causal learning with a fixed vocabulary. Causal graphical models separate intervention from association (Pearl, 2009; Spirtes et al., 2000); Ryall (2009) brings this machinery to strategy, modeling imitation as Bayesian inference over operating structures in which passive observation narrows candidates to an observational-equivalence class and experimentation may discriminate among survivors. Active-learning methods select interventions to split a represented equivalence class (He and Geng, 2008). These models are the benchmark for the paper's second learning tier: inference over structures expressible in a fixed vocabulary. The paper begins where they stop—when no expressible structure is adequate—and its nonidentification results extend the equivalence-class logic to the choice among vocabulary repairs: observationally equivalent worlds can demand different distinctions, so no truth-blind repair rule is uniformly correct.

Subjective rationality and self-confirming equilibrium. The back end follows Kalai and Lehrer (1993b) and Fudenberg and Levine (1993): beliefs correct on the path of play need not be correct off it, and subjectively optimal behavior against such beliefs can persist. Ryall (2003) showed that competitive advantage can rest entirely on rivals' self-confirming false theories. Those models fix the hypothesis space; convergence results in the rational-learning tradition require a grain of truth—the truth absolutely continuous with respect to beliefs (Kalai and Lehrer, 1993a). Under unawareness the grain-of-truth condition can fail at the level of language: the truth may be inexpressible. This paper supplies the missing dynamics. Statistical dissonance is the detector of language-level failure, awareness refinement restores the grain of truth on the path of play, and the fixed-language models are recovered exactly as the special case in which awareness never changes. The paper's absorbing configurations extend self-confirming equilibrium with two epistemic conditions—the language survives its own evidence, and no question is worth its price—and it is these conditions, not the strategic ones, that carry the new results.

Statement of contribution. Existing models show how theories guide learning, how Bayesian agents discriminate among represented causal structures, and how play can reveal previously unavailable actions. This paper models a different transition. An agent's causal language pools objectively different situations; strategic experience endogenously reveals an action-relevant failure of that pooling and thereby creates a localized open question; costly inquiry activates a fallible technology whose output is not measurable in the agent's prior awareness; only after that output is formulated does the agent construct a prior and undertake Bayesian judgment, under an explicit account of how much confidence the generating experience licenses. Because actions depend on theories, play can either create the contrasts that generate questions or suppress them—so the model also yields population-level sufficient conditions: for when shared false theories persist, and for when a diagnostic environment erases them.

3 The model

3.1 Agents, states, actions, and consequences

There are $n \geq 2$ *agents*, $N = \{1, \ldots, n\}$. Time is a single sequence of *periods* $t = 1, 2, \ldots$. In each period one state is drawn, every agent acts through a *standing policy* carried over from the previous period, and consequences are realized. Agents update beliefs every period, but revise their policy or their awareness only through the costly, deliberate steps specified below (assumption 3.4). Stretches of stable play are therefore not imposed but emerge: a policy persists until evidence moves it, so how long play stays fixed is an outcome—governed by agent-level parameters introduced below—rather than a modeling primitive.

In each period a *state* $s \in S$, with S finite, is drawn independently from a full-support distribution $\rho \in \Delta(S)$. States describe the strategically-relevant features of the environment that agents may face: customer configurations, input conditions, competitive circumstances, available production technologies, and so on. Their status in the agents' own representations is the subject of section 3.3.

Each period, agent i chooses an *action* a_i from a finite set A_i through the standing policy described below. An *action profile* is a list $a = (a_1, \ldots, a_n)$ where $A = \prod_i A_i$ is the set of all action profiles. Let $X = \prod_i X_i$ be the finite set of performance-related *consequences* with typical element $x = (x_1, \ldots, x_n)$. Agent i's consequence x_i is everything a period delivers to it—profit, sales, rival behavior, and any other observed events. Then, agent i's *payoff function* $\pi_i : X_i \to \mathbb{R}$ maps her consequences to payoffs.

3.2 The objective causal system

We take a very general view of how causality works in the world.

Definition 3.1 (Elementary mechanisms and the objective system)**.** *A finite set* Θ *of* elementary causal mechanisms *is given, each a map* $\theta : A \to \Delta(X)$*. The* objective assignment *is a function* $\tau : S \to \Theta$*; at state* s *under action profile* a *the resulting consequence distribution is* $\tau(s)(a) \in \Delta(X)$*, and this map from state–profile pairs to consequence distributions is the* objective causal system*. Two states are* causally equivalent *if they carry the same mechanism;* $\mathcal{P}^\tau$ *denotes the partition of* S *into causal-equivalence classes, and* $\operatorname{marg}_i \theta(a)$ *the* X_i*-marginal of* $\theta(a)$*.*

This causal reading is maintained throughout and does real work. The objective system is a response function, not a statistical summary: $\tau(s)(a)$ specifies the distribution of consequences that *would* result from every action profile at every state, including profiles never played. Agents' actions, together with the state, cause their consequences. A mechanism is the reduced form of whatever detailed model—a structural causal model of customer behavior, a market microstructure, a production process—underlies that relation; the paper imposes no structural-causal-model formalism, but examples may exhibit micro-foundations explicitly, as the running example's customer market does. Three consequences of the causal reading recur. First, theories, defined next, are interventional objects: they make predictions at unplayed profiles. Second, on-path data constrain $\tau(s)(a)$ only at visited state–profile pairs, so observationally equivalent systems with different counterfactual content are generic; that is the engine of the identification results. Third, the on-path distribution of consequences given an agent's category of states is a *selection* object—it mixes mechanisms according to which states co-occur with which play—and must never be conflated with any mechanism's own conditional.

3.3 Awareness, theories, and conjectures

Definition 3.2 (Awareness and expressible theories)**.** *Agent i's awareness partition is $\mathcal{P}_i \in \Pi(S)$, the set of partitions of S. Every consciously available object—theory, belief, conjecture, policy, question—must be measurable with respect to $\mathcal{P}_i$. The* expressible theories *are the cell-constant mechanism assignments*

$$M(\mathcal{P}_i) = \{m : \mathcal{P}_i \to \Theta\}, \tag{1}$$

with m predicting $x_i \sim \operatorname{marg}_i m(C)(a)$ at cell C under profile a. Partition $\mathcal{P}$ is sufficient *for an assignment τ' if τ' is constant on every cell of $\mathcal{P}$, equivalently if $\mathcal{P}$ refines $\mathcal{P}^{\tau'}$; the objective assignment is expressible for agent i if and only if $\mathcal{P}_i$ is sufficient for τ.*

An awareness cell is thus an implicit causal claim: the states it pools are alike in how actions produce consequences. An expressible theory makes the claim explicit by naming the mechanism. Restricting theories to *single* mechanism assignments—rather than arbitrary conditional distributions per cell—is substantive and deliberate. If any distribution were expressible, any within-cell mixture would be too, and no experience could reject the class: coarseness alone cannot produce a detectable failure in a saturated model space. The mechanism vocabulary Θ is known to all agents from the outset; what an agent may lack is the *distinction* that separates states carrying different mechanisms. Everything the paper calls insight is the production of such a distinction. Expansion of the mechanism vocabulary itself is a deeper dimension of awareness growth deliberately left outside this paper.

Agent i holds a belief $\mu_i \in \Delta(M(\mathcal{P}_i))$ over expressible theories—formally, the acting tier of a two-tier belief specified in section 5—and a conjecture $q_i : \mathcal{P}_i \to \Delta(A_{-i})$ about rival play, both $\mathcal{P}_i$-measurable. Theories and conjectures are different kinds of objects and are deliberately separated: a theory is a causal claim about the objective system; a conjecture is a strategic claim about what other agents will do. The diagnostic apparatus developed below tests theories, not conjectures.

Assumption 3.3 (Observation, retention, and the two records)**.** *At the start of each period agent i perceives its current awareness cell $C_{\mathcal{P}_i}(s)$; at the end it additionally observes the realized action profile a and its own consequence x_i. Each period is retained as a token. Two records are distinguished throughout. The* deliberative record *r_i is the multiset of tokens $(C_{\mathcal{P}_i}(s), a, x_i)$; it is the only record on which the agent's beliefs, tests, and decisions may depend. The* analyst record *$\bar{r}_i$ additionally carries the underlying state s of each token. Every object the paper attributes to the agent is a function of r_i; objects that depend on $\bar{r}_i$ (in particular the restorative correspondence and the warrant accounting of section 8) are the analyst's, and are so marked.*

Recodability. *When the agent acquires a new distinction—a refinement $\mathcal{P}'$ of its partition—it also acquires the capacity to classify each retained token under $\mathcal{P}'$, producing a recoded deliberative record. Before acquisition neither belief nor action can condition on that distinction. This capacity is a modeling idealization: it treats the retained particular as re-describable under any predicate the agent comes to* *possess, while denying the agent any pre-insight access to the predicate itself.*

Retention without recodability-in-advance is the minimal separation between experiencing a particular and possessing a projectible concept for it: the agent can remember "customers like that one" without commanding a predicate that distinguishes them, and the technology of section 6 operates on exactly this material. The idealization has a cost worth naming: recoding is treated as total and infallible once the predicate is held, whereas real reclassification of remembered particulars is partial and error-prone; section 8 isolates where this matters. Observability of the full action profile keeps the separation of causal from strategic error clean; a variant folding components of a into x_i changes bookkeeping only. The start-of-period perception of the current cell is what lets a cell-contingent policy be implemented; the state s itself is never perceived.

Assumption 3.4 (Standing policy, subjective optimality, and costly switching)**.** *Agent i carries a* standing policy $\sigma_i : \mathcal{P}_i \to A_i$ *from one period to the next. Given its current acting belief μ_i and conjecture q_i, the* belief-optimal *policy $\sigma_i^\star$ plays at each cell C*

$$\sigma_i^\star(C) \in \arg\max_{a_i \in A_i} \sum_{m \in M(\mathcal{P}_i)} \mu_i(m) \sum_{a_{-i} \in A_{-i}} q_i(a_{-i} \mid C)\, \mathbb{E}_{x \sim m(C)(a_i, a_{-i})}[\pi_i(x_i)]\,, \tag{2}$$

with associated subjective per-period value

$$U_i(\sigma \mid \mu_i, q_i) = \sum_{C \in \mathcal{P}_i} \rho(C) \sum_{m \in M(\mathcal{P}_i)} \mu_i(m) \sum_{a_{-i} \in A_{-i}} q_i(a_{-i} \mid C)\, \mathbb{E}_{x \sim m(C)(\sigma(C), a_{-i})}[\pi_i(x_i)]\,, \tag{3}$$

where $\rho(C) = \sum_{s \in C} \rho(s)$. Revising the standing policy is costly: agent i replaces σ_i by $\sigma_i^\star$ if and only if the per-period gain clears a switching hurdle $\kappa_i \geq 0$*, measured in per-period payoff units,*

$$U_i(\sigma_i^\star \mid \mu_i, q_i) - U_i(\sigma_i \mid \mu_i, q_i) > \kappa_i,$$

and otherwise carries σ_i unchanged. The inaction band $U_i(\sigma_i^\star) - U_i(\sigma_i) \leq \kappa_i$ is the source of policy inertia: beliefs may drift and the belief-optimal action may move, yet the standing policy holds until the gain from switching clears the hurdle. When dissonance (section 5) leaves μ_i undefined, $\sigma_i^\star$ is undefined and the standing policy is carried forward unchanged. Agents evaluate one period ahead and do not discount.

Policy inertia is thus an outcome, not an assumption: a standing policy persists until the belief-optimal action moves far enough to clear the switching hurdle κ_i, or until the language itself breaks down and must be repaired (section 5). How long play stays fixed is therefore governed by agent-level parameters—the switching hurdle here, and the tolerance for misfit introduced with the test of section 5—rather than by a fixed number of episodes.

3.4 On-path conditionals

Fix a stable stretch over which the standing profile $\sigma = (\sigma_1, \dots, \sigma_n)$ is constant. Each state s receives the deterministic profile $a(s) = (\sigma_j(C_{\mathcal{P}_j}(s)))_{j \in N}$. Agent i's *on-path conditional* at an observation pair (C, a) with $S(C, a) = \{s \in C : a(s) = a\} \neq \varnothing$ is

$$p_i^*(\cdot \mid C, a) = \sum_{s \in S(C,a)} \rho\big(s \mid S(C, a)\big) \ \mathrm{marg}_i \, \tau(s)(a). \tag{4}$$

This is the selection object flagged above: rivals with finer partitions condition on distinctions agent i cannot see, so the within-cell state mixture can vary with the profile. It is precisely this dependence—differentiated play by others inside one's own categories—that generates the contrasts on which the diagnostic apparatus feeds.

3.5 Modeling commitments

Five disciplines govern everything that follows; they are what distinguish a model of unawareness from a model of inattention or model selection. (i) Agents hold no prior over the outputs of the inquiry technology and never optimize over named future theories; pre-insight deliberation quantifies only over represented objects. (ii) Questions are generated by defects in the current representation, not drawn from a primitive list. (iii) Inquiry can fail, and its successes can be false; nothing reveals the objective assignment by construction. (iv) All valuations attached to inquiry are subjective, coarse, and possibly wrong; no objective payoff information leaks from behind the awareness frontier. (v) Evidence that generates a candidate representation is separated, in the formal accounting, from evidence that warrants it.

3.6 The running example

Example 1 (A four-state laboratory)**.** Two firms serve a unit market. States are customer configurations $S = \{s_{00}, s_{01}, s_{10}, s_{11}\}$, displayed as a 2×2 array with row coordinate r and column coordinate c; ρ is uniform. Actions are product architectures $A_i = \{0, 1\}$. The mechanism vocabulary is $\Theta = \{\theta^0, \theta^1, \theta^n\}$, with a demand parameter $\beta \in [0, 1]$: under θ^a ("architecture a is preferred"), if the firms differentiate, the firm offering a receives payoff 1 and its rival 0; if they pool on a, each receives $\frac{1}{2}$; if they pool on the other architecture, each receives $\beta/2$, the customer withholding $1 - \beta$ of the unit value. Under θ^n ("architecture is neutral"), each firm receives $\frac{1}{2}$ regardless. A consequence x_i is the pair (own payoff, observed action profile). The objective assignment is the *row law* $\tau(s_{rc}) = \theta^r$: the row coordinate is causally relevant, the column coordinate is not. Both firms begin with the coarse partition $\mathcal{P}_i^0 = \{S\}$, under which the expressible theories are the three constant assignments; the row law is not among them, because it is not constant on S. The baseline sets $\beta = 1$ (deterministic winner-take-all with market splitting); $\beta < 1$ returns in section 9. All mechanisms here are deterministic, so every test below operates in its exact, zero-tolerance instance; nothing in the general development relies on that.

4 Bayesian competence and its boundary

Within fixed awareness, learning is ordinary. Observations representable in the current language update μ_i by Bayes' rule; conjectures are updated empirically from observed play. If no agent's awareness ever changes, the model is a repeated interaction among subjectively rational Bayesian learners in the sense of Kalai and Lehrer (1993b) and Ryall (2003), and its stable configurations are self-confirming equilibria (Fudenberg and Levine, 1993). Everything new in this paper happens at the boundary of that layer, which two results locate.

Proposition 4.1 (No Bayesian emergence)**.** *Judged or ordinary conditioning of a prior $\mu_i \in \Delta(M(\mathcal{P}_i))$ on any observation $(C_{\mathcal{P}_i}(s), a, x_i)$ in the agent's observation algebra yields a posterior $\mu_i' \in \Delta(M(\mathcal{P}_i))$. No sequence of such updates makes a theory outside $M(\mathcal{P}_i)$ expressible or a distinction finer than $\mathcal{P}_i$ usable in a policy.*

Proof. Conditioning reweights a distribution on its domain and cannot enlarge the domain; every element of $M(\mathcal{P}_i)$ is cell-constant, so every posterior mixture, and every policy optimal against one, is $\mathcal{P}_i$-measurable. □

Remark 4.2 (The posterior is self-sealing)**.** *The stochastic setting yields a sharper point. Suppose every expressible theory is false: the agent's cell mixes mechanisms, so no cell-constant assignment matches the on-path conditionals. A full-support posterior does not signal this. By the logic of misspecified Bayesian learning (Berk, 1966), it concentrates on the expressible theories closest to the on-path conditionals in Kullback–Leibler divergence, and the agent becomes increasingly* confident *in a wrong theory. Posterior mass always sums to one; the inadequacy of the whole model class is not an event in the agent's algebra. Detecting that the language itself is broken therefore requires a faculty outside the posterior—the consistency test of the next section. The distinction matters for how the theory-based view should treat "updating": ranking the answers one can express and judging whether any expressible answer is adequate are different cognitive operations, and only the first is Bayesian.*

Example 1 (continued). Under the coarse partition, firm i's expressible theories are the constant assignments m^0, m^1, m^n. The row law τ is inexpressible: no reweighting of $\{m^0, m^1, m^n\}$, on any evidence, produces it (proposition 4.1). A firm that will later observe payoffs irreconcilable with all three theories will find its posterior undefined, not alarmed—the deterministic instance of remark 4.2, in which misspecification announces itself only because likelihoods hit zero exactly. With stochastic mechanisms, no announcement would occur at all.

5 Causal dissonance

5.1 Tests and active sets

Definition 5.1 (Sequential adequacy test, dissonance capacity, and active set)**.** *Fix an adequacy tolerance $\eta \geq 0$. Theory m is* adequate *at a visited pair (C, a) when* $\mathrm{TV}(p_i^*(\cdot \mid C, a), \mathrm{marg}_i\, m(C)(a)) \leq \eta$*, where* TV *is total-variation distance; under a standing profile call m η-*adequate *if it is adequate at every visited pair and η-*inadequate *if it is inadequate at some visited pair. The agent monitors adequacy* sequentially*, period by period. For each expressible theory m it maintains a nonnegative* evidence process *E_i^m, equal to 1 before any play and multiplied each period by a nonnegative factor for the observed token; the factor is built so that E_i^m is a supermartingale as long as m is adequate—its evidence against m does not grow in expectation—while at any pair where m is inadequate $\log E_i^m$ grows in expectation at a strictly positive rate, the information divergence between the on-path conditional and m's prediction. (A standard mixture construction over the adequacy region yields such a process; in the exact deterministic case $\eta = 0$ the factor is the likelihood ratio, so E_i^m becomes $+\infty$ the first period m assigns probability zero to a realized consequence.) Agent i carries a* dissonance capacity *$b_i > 0$ and* rejects *m, permanently, at the first period in which $\log E_i^m \geq b_i$. The* active set *$\widehat{M}_i(\mathcal{P}_i, r_i)$ collects the not-yet-rejected theories. Capacity is the agent's evidence threshold for abandoning a theory: by Ville's inequality the probability that an η-adequate theory is ever rejected is at most e^{-b_i}, so a high-capacity agent tolerates apparent misfit and rarely rejects on noise, while a low-capacity agent rejects on slight evidence and suffers more false alarms. Retrospective fits to a* fixed *record—the repair test of section 6 and the grouping test of section 7, which score a candidate against evidence already in hand rather than a stream—instead use the fixed-sample criterion*

$$\mathrm{TV}\big(\hat{p}_i(\cdot \mid C, a),\, \mathrm{marg}_i\, m(C)(a)\big) \leq \eta + w\big(n_i(C, a)\big) \qquad \textit{for every } (C, a) \textit{ with } n_i(C, a) \geq \bar{n}, \tag{5}$$

for a nonnegative, nonincreasing slack $w(\cdot) \to 0$ and minimum count $\bar{n} \geq 1$.

Lemma 5.2 (Test consistency)**.** *Fix a partition $\mathcal{P}_i$ and a standing profile, and suppose every theory is either η-adequate or strictly inadequate at each visited pair (none sits exactly at the adequacy boundary). Consider a record consisting of a finite prefix together with observations generated under that profile from some period on. Because rejection is permanent and the theories are finite, the active set is monotone and eventually constant; and almost surely:*

(a) every theory that is η-inadequate at a recurrent pair, and every theory carried past b_i by the frozen prefix alone, is eventually rejected; and

(b) every remaining theory—η-adequate at every recurrent pair and below b_i on frozen evidence—survives forever, except for a false rejection of probability at most e^{-b_i}.

Consequently dissonance arises almost surely when every expressible theory falls under (a); when some theory qualifies under (b), dissonance is avoided with probability at least $1 - e^{-b_i}$.

Proof. A recurrent pair is visited each period with a fixed positive probability under the i.i.d. full-support draw, hence infinitely often; a frozen pair is visited only in the finite prefix and adds at most a fixed constant to $\log E_i^m$. If m is inadequate at a recurrent pair, the evidence factor there has strictly positive expected log-growth (the information divergence), so by the strong law $\log E_i^m \to +\infty$ almost surely and m is rejected; likewise if the frozen constant alone reaches b_i. If m is η-adequate at every recurrent pair and its frozen evidence is below b_i, then E_i^m is a nonnegative supermartingale, so Ville's inequality gives $\Pr(\sup_t \log E_i^m \geq b_i) \leq e^{-b_i}$, and off that event m is never rejected. Permanence of rejection and finiteness of $M(\mathcal{P}_i)$ make the active set monotone and eventually constant; collecting the two cases gives the dissonance dichotomy. □

The frozen-evidence clause matters: a theory adequate under the current profile can be carried past b_i by legacy evidence from an earlier regime, so the active set tracks the whole retained record, not the current play alone. The deterministic case is the exact instance $\eta = 0$: an inadequate theory assigns probability zero to some consequence that occurs with positive on-path probability, so E_i^m becomes infinite the first time that consequence is realized—rejection is immediate and certain—while an adequate theory predicts every realized consequence with positive probability and is never rejected. Capacity plays no role here (any b_i gives the same verdicts) and the false-alarm probability e^{-b_i} is irrelevant, recovering the exact test.

Assumption 5.3 (Full-support marginals, stochastic case)**.** *In the stochastic case every mechanism marginal* $\operatorname{marg}_i \theta(a)$ *has full support on* X_i. *(The deterministic case, including the running example, is exempt and needs no such assumption.)*

The likelihood of a token (C, a, x_i) under theory m is $\operatorname{marg}_i m(C)(a)(x_i)$: the agent's subjective model treats the (C, a)-occurrence probability as theory-independent, so it cancels in Bayes' rule and updating requires neither ρ nor the within-cell state mixture. That this occurrence probability is objectively theory-informative—rivals' finer play correlates a_{-i} with the within-cell mechanism—is exactly the misspecification that feeds dissonance (remark 4.2); the agent does not model it.

Judged updating. Beliefs carry two tiers. The *reference posterior* $\bar{\mu}_i$ begins at a full-support prior on $M(\mathcal{P}_i)$ and updates by the token likelihoods above; it is never conditioned on the test. The *acting belief* μ_i is the reference posterior restricted and renormalized to the current active set, so that belief never rides on a rejected theory; when the active set is empty, the acting belief is undefined and the carry-forward clause of assumption 3.4 governs. Because the sequential test rejects permanently, the active set only shrinks, so the acting belief is well defined at every period until—if ever—the last expressible theory is rejected; assumption 5.3 keeps every reference likelihood positive, so the restriction is always to a positive-mass set. In the deterministic case rejection is a zero-likelihood event, so judged updating coincides with ordinary Bayesian conditioning. After a refinement, both tiers restart from the full-support bridge prior on the expanded space, from which judged updating resumes (section 8). Judged updating is the formal junction between the test and the posterior—between judgment and understanding (remark 4.2)—and it guarantees, in section 9, that surviving acting beliefs are supported on adequate theories.

5.2 Dissonance and exposure

Definition 5.4 (Causal dissonance)**.** *Agent i experiences* causal dissonance *at any period at which $\widehat{M}_i(\mathcal{P}_i, r_i) = \varnothing$: its sequential test has rejected every expressible theory. The expressible set does not disappear; no element of it accounts for the record. The immediate question has a determinate defect but not a determinate answer:* what unrepresented difference among the pooled states explains the heterogeneity of their consequences?

Dissonance is the model's rendering of a question for intelligence. It is localized (the failing cells are identified by the test), it is directed (an adequate repair must reconcile the record), and it is semantically open (no expressible object names the missing distinction). Whether dissonance ever occurs, however, depends on whether play makes the falsity of the language visible.

Definition 5.5 (Exposure and concealment)**.** *Under a standing profile, awareness $\mathcal{P}_i$ is* exposed *if every $m \in M(\mathcal{P}_i)$ is η-inadequate. A standing profile is* concealing *for agent i if some expressible theory is η-adequate under it while false as a claim about τ at some state its cells cover.*

Proposition 5.6 (Exposure and dissonance)**.** *Fix a standing profile and suppose no theory sits exactly at the adequacy boundary (lemma 5.2). If the record carries no frozen evidence that already rejects an η-adequate theory, then $\mathcal{P}_i$ is exposed if and only if agent i is dissonant from some period onward almost surely; when $\mathcal{P}_i$ is not exposed, dissonance is avoided with probability at least $1 - e^{-b_i}$. Absent that proviso, frozen legacy evidence can force dissonance even under a non-exposed profile.*

Proof. By lemma 5.2 the eventual active set consists of the theories η-adequate at every recurrent pair that lie below b_i on frozen evidence and escape false rejection. Under the no-legacy proviso frozen evidence rejects no η-adequate theory, so this set is empty almost surely iff no theory is η-adequate at every recurrent pair—exposure—while if such a theory exists it survives with probability at least $1 - e^{-b_i}$. If instead frozen evidence rejects every η-adequate theory, the active set is empty although the profile is not exposed. □

The proviso is not cosmetic: the dynamics of section 9 generate records whose early periods were played under coarser partitions and different profiles, so the frozen evidence is real, and the characterization of the corrective forces below is stated for the terminal regime where the proviso holds by construction (theorem 9.5(d)).

Exposure has two sources, and the distinction organizes much of what follows. Heterogeneity inside a cell surfaces either because *other agents play differently across the pooled states*—strategic differentiation inside one's categories, which changes the visited profiles and the induced mixtures—or because *the environment punishes uniform play across causally different states in an observable way*. Both reappear as the corrective forces in section 9.

Example 1 (continued). Let $\beta = 1$ and suppose the firms' priors over the constant theories are heterogeneous: firm 1 considers θ^0 more likely than θ^1, and firm 2 the reverse. A direct computation shows that, against any conjecture about the rival, offering architecture 0 is subjectively strictly dominant for firm 1 and architecture 1 for firm 2 whenever $\mu_i(m^a) > (2-\beta)\,\mu_i(m^{a'})$; at $\beta = 1$ this is simple belief asymmetry. The firms therefore commit to differentiated constant policies. Set $T = 2$ and condition on the discovery generation serving the two diagonal states s_{00} and s_{11}—a positive-probability event under uniform ρ; the appendix records how the leverage below scales with the realized composition. With differentiated actions, the winner-take-all payoffs reveal the objectively preferred architecture at each state: firm 1 wins at s_{00} (payoffs $1, 0$) and loses at s_{11} (payoffs $0, 1$). No constant assignment is consistent with both episodes—m^0 fails at s_{11}, m^1 at s_{00}, and m^n predicts one-half at both—so each firm's active set empties: causal dissonance, in its exact deterministic form. Had the firms instead pooled on the same architecture at both states, each would have received one-half at every episode, an outcome consistent with all three constant theories: the pooled profile is concealing, and coarse awareness would have survived its own evidence indefinitely. Belief heterogeneity, by inducing differentiated play, is what made the shared language's failure visible.

6 Inquiry: restoration, direction, and fallibility

Dissonance poses a question the current language cannot answer. This section characterizes what a successful answer must accomplish, shows that the requirement directs inquiry without determining its output, and establishes that any inquiry process meeting the paper's disciplines must be fallible.

6.1 Restorative refinements

For $\mathcal{P}, \mathcal{P}' \in \Pi(S)$, write $\mathcal{P} \prec \mathcal{P}'$ when $\mathcal{P}'$ strictly refines $\mathcal{P}$. The restorative correspondence is an *analyst-level* object: deciding whether a candidate refinement $\mathcal{P}'$ restores expressibility requires recoding tokens to $\mathcal{P}'$-cells, which uses the analyst record $\bar{r}_i$ of assumption 3.3. It is defined here for the analysis of what a successful answer must accomplish; section 7 builds the agent's own deliberative valuation separately.

Definition 6.1 (Restorative refinement)**.** *Fix a candidate refinement $\mathcal{P}_i \prec \mathcal{P}'$ and recode the analyst record $\bar{r}_i$ to the deliberative record $r_i^{\mathcal{P}'}$ on $\mathcal{P}'$. Define the* repair-consistent active set

$$\widehat{M}_i^R(\mathcal{P}', \bar{r}_i) = \left\{ m' \in M(\mathcal{P}') : m' \text{ satisfies equation (6)} \right\},$$

where the repair test *is*

$$\mathrm{TV}\big(\hat{p}_i(\cdot \mid C', a), \mathrm{marg}_i\, m'(C')(a)\big) \le \eta + w\big(n_i(C', a)\big) \quad \textit{for every } (C', a) \textit{ with } n_i(C', a) > 0. \tag{6}$$

Here the counts and empirical conditionals are computed from $r_i^{\mathcal{P}'}$*. The refinement* $\mathcal{P}'$ *is* restorative *at* $(\mathcal{P}_i, \bar{r}_i)$ *if* $\widehat{M}_i^R(\mathcal{P}', \bar{r}_i) \neq \varnothing$*. It is* minimally restorative *if no restorative* $\mathcal{P}''$ *satisfies* $\mathcal{P}_i \prec \mathcal{P}'' \prec \mathcal{P}'$*. Write* $\mathcal{R}(\mathcal{P}_i, \bar{r}_i)$ *for the set of minimally restorative refinements.*

The repair test deliberately applies to every nonempty recoded cell–profile pair. The count floor $\bar{n}$ governs when a recoded pair carries enough evidence for the repair test to bind; it does not permit a proposed repair to erase retained evidence by splitting a tested cell into fragments below that floor. Every nonempty fragment remains subject to equation (6), while the weakly larger finite-sample slack $w(n)$ at small n preserves the intended caution about sparse evidence. Refinement can therefore reconcile observations but cannot delete their constraints. In the exact deterministic case $\eta = 0$ and $w \equiv 0$, the repair test is exact consistency at every observed pair.

Proposition 6.2 (Cellwise coherence)**.** *For a candidate refinement* $\mathcal{P}_i \prec \mathcal{P}'$*, define*

$$\Theta_i^R(C' \mid \bar{r}_i) = \left\{ \theta \in \Theta : \begin{array}{r} \mathrm{TV}\big(\hat{p}_i(\cdot \mid C', a), \mathrm{marg}_i\, \theta(a)\big) \le \eta + w\big(n_i(C', a)\big) \\ \textit{for every } a \textit{ with } n_i(C', a) > 0 \end{array} \right\}.$$

Call C' repair-coherent *if* $\Theta_i^R(C' \mid \bar{r}_i) \neq \varnothing$*. Then* $\mathcal{P}'$ *is restorative if and only if every cell of* $\mathcal{P}'$ *is repair-coherent.*

Proof. Because $M(\mathcal{P}') = \Theta^{\mathcal{P}'}$ and every constraint in equation (6) involves only the mechanism assigned to one cell,

$$\widehat{M}_i^R(\mathcal{P}', \bar{r}_i) = \prod_{C' \in \mathcal{P}'} \Theta_i^R(C' \mid \bar{r}_i).$$

This product is nonempty exactly when every factor is nonempty. □

Coherence is the general form of a conflict structure. Token sets are *mutually incoherent* when no single mechanism accommodates their union; every restorative refinement must separate mutually incoherent sets, since a cell containing both would be incoherent. In the deterministic running example incoherence is a binary relation between labeled episodes and restoration is a graph-coloring problem; in general, a mixture of three mechanisms may defeat every single mechanism even though each pair of tokens is individually accommodatable, so restoration is a hypergraph coloring. Two further structural facts matter. First, minimal restorations, when they exist, are typically multiple—as the example shows vividly. Second, existence itself is guaranteed only in the deterministic case.

Remark 6.3 (Irreparable records)**.** *With deterministic mechanisms, every dissonant record is repairable. Under the exact calibration* $\eta = 0$ *and* $w \equiv 0$*, consider the singleton partition of* S*. At each state* s

and each observed profile a, every retained consequence was generated by the deterministic mechanism $\tau(s)$; hence $\tau(s)$ fits all observations recoded to $\{s\}$ exactly. The objective assignment therefore belongs to the repair-consistent active set on the singleton partition. If the current partition is already the singleton partition, dissonance is impossible; otherwise finiteness of $\Pi(S)$ ensures a minimally restorative refinement.

With stochastic mechanisms this guarantee fails. Multiple observations at a single state may yield empirical conditionals for which no one mechanism satisfies equation (6) across the observed profiles. Because no refinement can split a state, such a record has $\mathcal{R}(\mathcal{P}_i, \bar{r}_i) = \varnothing$ and is irreparable*: dissonance names a defect that no expressible refinement can repair. Irreparability is an analyst-level property: it is a fact about $\bar{r}_i$, and the agent, lacking the state labels, cannot in general tell an irreparable record from a repairable one—this is precisely the agent's inability to distinguish causal heterogeneity from mechanism noise, developed in section 7. When the record is irreparable the technology fails surely (definition 6.4); the resulting behavior is governed by the deliberative valuation and failure-discouragement of section 7. If the deliberative record still admits a positive-leverage coherent grouping, the agent may elect a doomed inquiry until its achievability weight decays; if no coherent grouping prices the question, it declines immediately. Neither response requires the agent to recognize analyst-level irreparability. Under recurring visitation irreparability is typically transient, but a record whose critical pairs stop being visited can remain irreparable forever, a possibility section 9 classifies within dissonant stasis.*

6.2 The technology

Definition 6.4 (Representation-producing technology)**.** *A representation-producing technology for agent i is a possibly randomized map ϕ_i from the analyst record $(\mathcal{P}_i, \bar{r}_i)$ to $\{\mathcal{P}_i\} \cup \mathcal{R}(\mathcal{P}_i, \bar{r}_i)$: it fails, returning the current partition, or succeeds, producing a minimally restorative refinement. When $\mathcal{R}(\mathcal{P}_i, \bar{r}_i) = \varnothing$, the range is the singleton $\{\mathcal{P}_i\}$ and the technology fails surely. It is* truth-blind*: its output distribution depends only on $(\mathcal{P}_i, \bar{r}_i)$, never directly on τ. The technology consumes the analyst record because producing a distinction operates on retained particulars in their full concreteness; what the agent lacks pre-insight is not the particulars but the predicate that organizes them, and truth-blindness records that even this concrete operation cannot read off the objective law. A complexity ordering on refinements—the analyst's model of which conceptual products the agent's comparison capacities can generate—may restrict the range further, provided it leaves the admissible subset of $\mathcal{R}$ nonempty whenever $\mathcal{R}$ is; it is a primitive of the technology, not an object of the agent's optimization.*

Definition 6.5 (Truth-blind formulation rule)**.** *After successful production of $\mathcal{P}'_i$, let $r_i^{\mathcal{P}'_i}$ denote the deliberative record recoded to that partition. At this point the agent can compute from $r_i^{\mathcal{P}'_i}$ the same repair-consistent active set that the analyst writes as $\widehat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)$. A* formulation rule *is a possibly randomized kernel*

$$\psi_i\Big(\cdot \mid \mathcal{P}'_i, r_i^{\mathcal{P}'_i}\Big) \in \Delta\Big(\widehat{M}_i^R(\mathcal{P}'_i, \bar{r}_i)\Big)\,.$$

It is record-measurable and truth-blind *if its conditional distribution depends only on the displayed* *arguments and on internal randomness independent of the objective assignment, state draws, consequence draws, and production randomness. A* truth-blind inquiry-and-formulation rule *is a pair* (ϕ_i, ψ_i) *satisfying these restrictions. Any randomized post-refinement prior-selection rule is subject to the same record-measurability and truth-blindness requirement.*

The technology is the paper's stated representational primitive. No formal model derives a semantically new concept from nothing; the contribution is to show how experience *directs* a limited, fallible generative capacity and why that capacity does not amount to a prior over future theories. The output, and even the set of possible outputs, is not a pre-insight subjective menu: the agent cannot name a candidate refinement, assign it a probability, or condition a policy on it before production succeeds (section 3.5).

6.3 Nonidentification and the price of robustness

Theorem 6.6 (Observational nonidentification)**.** *Fix common initial epistemic states, common record-measurable behavioral and tie-breaking rules, common truth-blind inquiry-and-formulation rules, and a finite generation horizon G. Let $\tau \neq \tau'$ be objective assignments whose* joint *consequence laws agree at every state–profile pair reachable through G under either world from those initial conditions:*

$$\tau(s)(a) = \tau'(s)(a) \in \Delta(X) \qquad \text{at every such } (s, a).$$

The two worlds admit a coupling under which, through G, their state draws, full consequence profiles, analyst and deliberative records, partition paths, beliefs, conjectures, committed policies, and production-and-formulation outputs coincide pathwise. If the equality holds at every pair reachable at any finite horizon, the coupling extends to the entire process.

Consequently, no truth-blind inquiry-and-formulation rule can produce an objectively correct formulation with probability one in both worlds. More generally, the same conclusion holds for any success criterion whose sets of realizable outputs are disjoint across the two worlds.

Proof. Use common state draws, common independent random seeds for production, formulation, bridge-prior selection, and tie-breaking, and an identity coupling of the joint consequence draws at every reachable pair. Proceed by induction over episodes and generations. Identical histories imply identical current partitions and committed policies, hence the same action profile at the common state. The pair is reachable, so the two objective assignments give it the same joint consequence distribution and the coupled consequence profiles coincide. Both analyst and deliberative records therefore remain identical. Record-measurability and truth-blindness then give identical belief updates, conjectures, policies, refinements, and formulations, closing the induction through G; the all-horizons claim follows by countability.

Thus every truth-blind rule has the same output distribution in the two worlds. Because $\tau \neq \tau'$, their sets of objectively correct formulations are disjoint, so that common distribution cannot assign probability one to both sets. The same argument applies to any other disjoint pair of success sets. $\square$

Proposition 6.7 (Robust sufficiency)**.** *A partition sufficient for every assignment in a candidate set $\mathcal{T}$ must refine $\bigvee_{\tau' \in \mathcal{T}} \mathcal{P}^{\tau'}$, the join of the candidates' causal-equivalence partitions. A representation guaranteed to accommodate all observationally equivalent worlds is accordingly finer than any single world requires; a technology that economizes on distinctions must accept fallibility.*

Proof. Sufficiency for τ' is refinement of $\mathcal{P}^{\tau'}$ (definition 3.2); refinement of each candidate's partition is refinement of their join. $\square$

Example 1 (continued). After the dissonant generation, the record contains two labeled episodes: architecture 0 objectively preferred at s_{00}, architecture 1 at s_{11}. Of the fifteen partitions of four states, exactly four are minimally restorative—each must separate s_{00} from s_{11}: the row partition $\{\{s_{00}, s_{01}\}, \{s_{10}, s_{11}\}\}$, the column partition $\{\{s_{00}, s_{10}\}, \{s_{01}, s_{11}\}\}$, and the two partitions isolating one labeled state. The record directs inquiry (eleven partitions are excluded) without determining it (four remain). A natural complexity ordering counts the coordinates needed to describe a refinement: rows and columns are one-coordinate distinctions, the isolating partitions require two. A factor-seeking technology thus produces the row or the column refinement—and here direction runs out. Define the *column world* $\tau'(s_{rc}) = \theta^c$. Through the discovery horizon and conditional on the diagonal-state event, it agrees with the row world on the full joint mechanism at both visited states, so the two worlds generate the discovery record and every truth-blind production-and-formulation output with the same probability by the same coupling argument, conditional on the diagonal-state event. No truth-blind rule can formulate correctly in both, a randomized rule succeeds in the worst case with probability at most one-half, and by proposition 6.7 a partition sufficient for both worlds must be the full partition into singletons: robustness costs every distinction the language can draw, which is exactly what a minimal technology economizes on. The example also locates the failure mode of each residual candidate: whichever factor refinement arrives, its unique consistent theory extrapolates the wrong prediction to the two unlabeled states in one of the two worlds.

7 The economics of inquiry

Whether to prosecute a question must itself be a decision, and it must be priced at exactly the moment ordinary decision theory is silent: at dissonance the posterior is undefined (definition 5.4), and the candidate answers are not represented objects (definition 6.4). This section builds the valuation from what the agent actually has—its deliberative record—and prices the rest through the success weight. The resulting statistic is a genuine estimate, not a guarantee: the agent guesses at the worth of a distinction it cannot yet name, and the model keeps that epistemic honesty rather than smuggling in the analyst's restorative correspondence.

Assumption 7.1 (Costly, elective, dissonance-triggered production)**.** *The technology ϕ_i operates only when agent i experiences dissonance and elects inquiry, at cost $\gamma_i > 0$ in payoff units. Let k_i count agent i's consecutive failed elections, reset to zero on any success. Agent i holds a* delabeled achievability weight $\lambda_i^{(k)} \in (0, 1]$, *nonincreasing in k with $\lambda_i^{(k)} \to 0$: a subjective probability that an elected episode produces a refinement* delivering the estimated value below*, defined on that event alone and not on the identities of possible outputs. On failure or abstention the committed policy is carried forward. Awareness is cumulative: acquired distinctions are never lost, though theories formulated on them may later be rejected.*

The weight is awareness of unawareness in delabeled form—the agent can believe that inquiry may yield a usable distinction, and grow less hopeful as attempts fail, without ever describing a candidate distinction—a construct with precedent in beliefs about unpredictable novelty (Schipper, 2025; Karni and Vierø, 2017). Its decay is the answerability meta-prior doing formal work: repeated failure is evidence that the question may be unanswerable (the record irreparable, the repertoire inadequate), and a rational inquirer eventually abandons it. That the weight prices *achievability*—not mere success—absorbs the gap between the agent's estimate and what an arriving refinement can actually deliver, a gap that is zero under the pointwise-transparency condition below and may be positive otherwise.

Definition 7.2 (Deliberative groupings and coherent classes)**.** *A* deliberative grouping *G of the record r_i is a partition of its tokens. For a group $g \in G$ and observed profile a, let*

$$n_g(a) = \big|\{t \in g : a(t) = a\}\big|$$

and, when $n_g(a) > 0$, let $\hat{p}_g(\cdot \mid a)$ be the empirical distribution of the consequences $x_i(t)$ among those tokens. Define

$$\Theta_i(g) = \big\{\theta \in \Theta : \mathrm{TV}\big(\hat{p}_g(\cdot \mid a), \mathrm{marg}_i\, \theta(a)\big) \le \eta + w\big(n_g(a)\big) \text{ for every } a \text{ with } n_g(a) \ge \bar{n}\big\} .$$

The group g is coherent *if $\Theta_i(g) \neq \varnothing$, and the grouping G is coherent if every one of its groups is coherent. Write $\mathcal{G}_i(r_i)$ for the finite family of coherent groupings of r_i.*

Coherence, $\Theta_i(g)$, and $\mathcal{G}_i(r_i)$ depend only on the deliberative record—on observed actions and own consequences—together with the known mechanism vocabulary. They never use state labels or candidate refinements. The family $\mathcal{G}_i(r_i)$ may be empty under stochastic sampling: full support guarantees positive likelihood, not finite-sample passage of the consistency test. An empty family means only that the retained record supplies no coherent action-contingent decomposition the agent can price; it does not tell the agent whether an analyst-level restorative refinement exists. A coherent grouping is the agent's hypothesis about which remembered particulars share a mechanism. It is not a partition of S, and the agent cannot verify before insight that any refinement of its awareness would realize it.

Definition 7.3 (Deliberative leverage). *Given a coherent grouping G, let $g(t)$ be the group of token t and $\sigma_i(t)$ the carried-forward action at t's current awareness cell. The* per-token robust gain *of proposing action a_i for a group is*

$$\Delta_i(t, a_i) = \min_{a_{-i} \in A_{-i}} \min_{\theta \in \Theta_i(g(t))} \Big(\mathbb{E}_{x \sim \theta(a_i, a_{-i})}[\pi_i(x_i)] - \mathbb{E}_{x \sim \theta(\sigma_i(t), a_{-i})}[\pi_i(x_i)] \Big),$$

the worst case over rival responses and over the mechanisms the group's record leaves open. For $G \in \mathcal{G}_i(r_i)$, define its aggregate robust gain by

$$L_i(G; r_i) = \sum_{g \in G} \max_{a_i \in A_i} \sum_{t \in g} \Delta_i(t, a_i).$$

The deliberative leverage *is*

$$\widehat{B}_i(r_i) = \begin{cases} \dfrac{T}{|r_i|} \max\big(\{0\} \cup \{L_i(G; r_i) : G \in \mathcal{G}_i(r_i)\}\big), & \widehat{M}_i(\mathcal{P}_i, r_i) = \varnothing, \\ 0, & \widehat{M}_i(\mathcal{P}_i, r_i) \neq \varnothing. \end{cases} \tag{7}$$

The premise is that the empirical token distribution *recurs, so the per-generation estimate is T times average per-token robust gain. The outside option* 0 *represents declining to attribute actionable value to any available grouping; it also makes leverage zero when $\mathcal{G}_i(r_i) = \varnothing$.*

Proposition 7.4 (Participation). *$\widehat{B}_i(r_i)$ is well defined, nonnegative, bounded by $T(\max \pi_i - \min \pi_i)$, and computable from the deliberative record, the carried-forward policy, and the known structure (Θ, π_i). At a dissonant record, electing inquiry has represented value $\lambda_i^{(k_i)} \widehat{B}_i(r_i) - \gamma_i$, so agent i elects if and only if*

$$\lambda_i^{(k_i)} \widehat{B}_i(r_i) \geq \gamma_i.$$

Moreover, at a dissonant record, $\widehat{B}_i(r_i) > 0$ if and only if some coherent grouping has positive aggregate robust gain:

$$L_i(G; r_i) > 0 \qquad \text{for some } G \in \mathcal{G}_i(r_i).$$

Thus a question is priced positively only when some coherent reading of the record supports a group-contingent action proposal that robustly improves on the carried-forward policy in the aggregate.

Proof. The record is finite, so it has finitely many groupings. For every coherent group, $\Theta_i(g)$ is a nonempty finite set, and all action and consequence spaces are finite. Hence every max–min expression and every $L_i(G; r_i)$ is well defined. The set whose maximum appears in equation (7) always contains 0, including when $\mathcal{G}_i(r_i) = \varnothing$, proving existence and nonnegativity. Writing $\Delta\pi_i = \max \pi_i - \min \pi_i$, each $\Delta_i(t, a_i) \leq \Delta\pi_i$. Every grouping contains exactly $|r_i|$ tokens, so $L_i(G; r_i) \leq |r_i| \Delta\pi_i$ and therefore $\widehat{B}_i(r_i) \leq T\Delta\pi_i$. Computability follows by finite enumeration using only r_i, the carried-forward policy, Θ, and π_i. The election rule is the achievabilityweighted represented gain net of cost. Finally, because the outside option is zero, the maximum in equation (7) is strictly positive exactly when some coherent grouping has strictly positive aggregate robust gain. □

The participation rule is a coherent non-Bayesian criterion operating precisely where Bayes is undefined, and it respects every discipline of section 3.5: $\widehat{B}_i$ quantifies over the agent's own remembered tokens and the groupings it can articulate—"situations like the ones I lost will recur, and a distinction that sorted them would let me act on them"—never over named future representations. The estimate can be wrong, and honestly so: the agent does not know whether a distinction realizing its best grouping exists, which is why $\lambda_i^{(k)}$ prices achievability. Two comparative properties organize the applications. *Action separation, not information*: the value of a question tracks whether resolving it could change behavior, not how much residual uncertainty remains; an agent for whom no coherent grouping yields positive aggregate robust gain declines to inquire. *Rational cessation*: an agent for whose record no coherent grouping yields positive aggregate robust gain has $\widehat{B}_i = 0$, so sustained success can extinguish inquiry even though awareness stays incomplete; and, by the decay of $\lambda_i^{(k)}$, so can sustained failure—a true, valuable, repairable question can be rationally abandoned.

Remark 7.5 (What determinism certifies—and what it does not)**.** *Determinism supplies a limited but useful separability certificate. Two tokens sharing the same observed pair (C, a) but differing in consequence cannot have arisen from the same state, because a state and action profile fix the consequence. Any restorative refinement must therefore separate such a certified conflict. Determinism alone does not, however, make every deliberative grouping realizable. Tokens observed under different action profiles can be grouped coherently even when their underlying particulars cannot be separated in the proposed way. In general, $\widehat{B}_i$ therefore remains a deliberative estimate even with deterministic mechanisms.*

A stronger, record-level condition restores exactness. In the deterministic zero-tolerance case with $\bar{n} = 1$ and $w \equiv 0$, call a record pointwise transparent *if $\Theta_i(\{t\})$ is a singleton for every retained token t. Then every coherent group contains only tokens carrying the same pinned mechanism, while every restorative refinement separates tokens carrying different pinned mechanisms. Tokens with the same pin and the same current awareness cell also have the same robust action comparison; tokens in different current cells remain separately actionable because every new partition refines the current one. Consequently every maximizing group-contingent proposal is implementable under any restorative refinement, and the deliberative leverage is exact and repair-invariant. The running example's differentiated discovery record is pointwise transparent, so its leverage calculation has precisely this property.*

Under stochastic mechanisms, differing consequences at one (C, a)-pair may reflect mechanism noise rather than distinct states, and even a coherent grouping need not be realizable. There $\widehat{B}_i$ is genuinely an estimate, and the achievability weight prices the possibility that no arriving distinction delivers it. Counterfactual transparency—a small $\Theta_i(g)$, ideally a singleton—makes the estimate sharper; opacity enlarges $\Theta_i(g)$, lowers robust leverage, and can rationally suppress inquiry.

Example 1 (continued). At the discovery record ($T = 2$, diagonal states served), each firm holds two tokens sharing the profile $(0, 1)$ but differing in payoff: firm 1 won at s_{00} and lost at s_{11}, firm 2 the reverse. Because mechanisms are deterministic, the conflicting payoffs *certify* the two tokens as distinct states (remark 7.5), and the agent's best coherent grouping splits them: the winning token pins its architecture as preferred (current action stays optimal, contributing zero), the losing token pins the other architecture (switching to it gains $\frac{1}{2}$ against every rival action—a differentiated loss becomes a pooled half, or a pooled half a differentiated win). The two tokens average a per-token gain of $(\frac{1}{2} + 0)/2 = \frac{1}{4}$, so $\widehat{B}_i = (T/|r_i|) \sum_t \Delta = (2/2) \cdot \frac{1}{2} = \frac{1}{2}$ for both firms—identical leverage from opposite experiences—so each elects iff $\lambda_i^{(0)} \widehat{B}_i \geq \gamma_i$, i.e. $\lambda_i^{(0)}/2 \geq \gamma_i$. The valuation used no conjecture about the rival and no reference to which refinement might arrive; transparency (winner-take-all) makes each $\Theta_i(g)$ a singleton, so the deliberative estimate is exact. For rational cessation and its converse, append a validation generation in which firm 1 has acquired the row refinement (certain of the true assignment) while firm 2 has acquired the column refinement (certain of the false one): their policies differ at the two off-diagonal states, firm 1 wins both, and the labels revealed there falsify firm 2's column theory. Firm 1's record then contains no coherent group favoring a switch, so $\widehat{B}_1 = 0$: the firm holding the truth rationally never inquires again, and its awareness stays permanently incomplete. Firm 2's record now carries losing off-diagonal tokens its column policy misplays, so $\widehat{B}_2 > 0$: defeat concentrates the evidence, the dissonance, and the incentive in the losing firm, which elects whenever $\lambda_2^{(k_2)} \widehat{B}_2 \geq \gamma_2$ and, upon success, reaches complete awareness in one step (both column cells must split). The winner/loser asymmetry—$\widehat{B}_1 = 0 < \widehat{B}_2$—is the durable point; the appendix records the exact value of $\widehat{B}_2$.

8 The pivot: formulation, warrant, and the return to Bayes

Successful production yields a refinement $\mathcal{P}_i'$; the truth-blind formulation rule selects $m' \in \widehat{M}_i^R(\mathcal{P}_i', \bar{r}_i)$ from the repair-consistent theories on the recoded record; and the agent forms a full-support *bridge prior* $\mu_i^+ \in \Delta(M(\mathcal{P}_i'))$, from which ordinary updating resumes. Nothing in Bayesian coherence determines μ_i^+ from the pre-dissonance belief—the domains differ—and this nonuniqueness is not a defect to be repaired but the boundary between insight and judgment. Renormalizing μ_i^+ on the active set yields what we call *representation-relative coherence*: the confidence the new language awards its own theories. The question of this section is how much of that confidence the evidence actually licenses.

The conjecture on the refined partition is initialized from the recoded empirical frequencies of observed rival actions wherever the record contains observations, with a fixed default on previously unvisited cells, and is updated empirically thereafter. Thus both inputs needed for the next committed policy—the bridge belief and the strategic conjecture—are defined on the expanded language only after the refinement has arrived.

Definition 8.1 (Reflective assessment, warrant, and the gap)**.** *A reflective assessment problem is a pair $(\mathcal{T}, \nu)$: a finite set of candidate objective assignments and a full-support reflective prior. It is an* analyst-level *construct: conditioning on the events below uses the analyst record and the inquiry history. Let h_i denote the pre-production inquiry history, including earlier elections and failures. Given $(\bar{r}_i, h_i)$ together with a realized successful production-and-formulation output $(\mathcal{P}_i', m')$, the* selection-adjusted warrant *W of $(\mathcal{P}_i', m')$ is the ν-posterior probability that m' is objectively correct—that $\tau'(s) = m'(C_{\mathcal{P}_i'}(s))$ for all s—conditional on that record* and *history and on the realized output. The* warrant gap *is the representation-relative coherence probability of m' minus its warrant W.*

Theorem 8.2 (The insight is not evidence)**.** *Condition on the analyst record $\bar{r}_i$ and the pre-production inquiry history h_i. Under a record-measurable, truth-blind inquiry-and-formulation rule, conditioning additionally on a particular successful output $(\mathcal{P}_i', m')$ does not change reflective odds:*

$$W = \nu\big(m' \text{ is objectively correct} \mid \bar{r}_i, h_i\big) .$$

Post-restoration fit is guaranteed by the restoration property, so passing the repair test carries no information beyond the record either. More generally, among candidates in $\mathcal{T}$ that induce the same likelihood for $(\bar{r}_i, h_i)$, reflective prior odds are preserved exactly.

Proof. For $\tau' \in \mathcal{T}$, the posterior conditional on the record, history, and realized output is proportional to

$$\nu(\tau') \Pr_{\tau'}(\bar{r}_i, h_i)\, K_i\big(\mathcal{P}_i', m' \mid \mathcal{P}_i, \bar{r}_i, h_i\big) ,$$

where K_i is the conditional kernel generated by election, production, and formulation. Election is determined by the deliberative record and failure counter, while production and formulation are truth-blind. Consequently the last factor depends on the displayed record and history but not on τ', and cancels from posterior odds. Passing the repair test is a deterministic implication of the record and realized output and therefore contributes no additional likelihood factor. If the record-and-history likelihoods are also equal across two candidates, their posterior odds equal their prior odds. □

The theorem locates the evidential boundary: everything evidential is in the record; nothing evidential is in the arrival of the insight or in its retrospective success at organizing the experience that produced it. The conditioning must be at the analyst level, because the election and the success set are functions of the state-labeled record; were one to condition only on the deliberative record, the distribution of the hidden states would itself be τ'-dependent and the cancellation would fail—the insight would spuriously appear to be evidence. The practical force is a prohibition on double counting: the same experience that directed production cannot also confirm the resulting theory, because any restorative output fits it by construction. Truth blindness is the polar case; were production correlated with the objective assignment—social transmission from an informed source—the arrival would itself be evidence, and the adjustment could move confidence in either direction.

Corollary 8.3 (Self-audit requires sufficiency for the candidates)**.** *The assessment $(\mathcal{T}, \nu)$ is expressible by agent i only if $\mathcal{P}_i$ is sufficient for every $\tau' \in \mathcal{T}$, hence only if $\mathcal{P}_i$ refines the join of proposition 6.7. An agent holding a minimal restoration therefore generally cannot formulate the comparison that measures its own warrant gap: the overconfidence is invisible from within the representation that produces it. Short of that refinement, warrant must be supplied externally—by fresh evidence whose likelihoods differ across candidates, by an assessor with broader awareness, or by a future self.*

Proof. Entertaining τ' as a hypothesis requires expressing it, which is sufficiency (definition 3.2); sufficiency for every candidate is refinement of the join. □

The pivot is now complete, and it is the paper's fulcrum. Dissonance broke the agent's Bayesian competence; inquiry produced a distinction; formulation and the bridge prior rebuilt a probability space; and judgment—fresh evidence with differential likelihoods, honestly accounted—resumes as ordinary updating on the expanded language. The agent has returned to the Bayesian world carrying a warrant gap of known location and, by corollary 8.3, generally unknowable (to it) size.

Example 1 (continued). Suppose firm i's technology produces the column refinement. The recoded record pins the unique consistent theory: architecture 0 on the first column, 1 on the second. Representation-relative coherence assigns it probability one. Now conduct the reflective assessment with $\mathcal{T} = \{\text{row world}, \text{column world}\}$ and a symmetric prior. The two worlds assign the realized diagonal-only record equal probability by the same coupling argument conditional on the diagonal-state event, and the technology is truth-blind, so by theorem 8.2 the reflective posterior equals the prior: $W = \frac{1}{2}$, and the warrant gap is exactly one-half. The same gap afflicts the firm that acquires the *row* refinement: its coherence in the true assignment is also probability one, and its warrant is also one-half. Unwarranted certainty is not a symptom of error—early confidence outruns the evidence even when it happens to be right, and what separates the two firms is the objective law, which neither observes. By corollary 8.3, neither firm can conduct this assessment itself: expressing both candidate worlds requires the full partition into singletons. One fresh differentiated observation at an off-diagonal state closes the gap completely—the two worlds predict opposite labels there—while pooled observations, in any number, close nothing. Warrant, like leverage, is purchased with differentiated action.

9 Strategic dynamics: when everyone lives in a Bayesian world

The chain now runs inside a population. Theories determine committed policies; policies determine which conditionals are visited; visited conditionals determine dissonance and leverage;

and refinements feed back into theories. This section characterizes where the process comes to rest. The object of interest is not a strategic equilibrium but an epistemic condition: a configuration of the population in which *every agent permanently inhabits a Bayesian world*—models fit, questions are dead, and no represented calculation prices further inquiry above its cost.

Definition 9.1 (Self-confirming unawareness equilibrium)**.** *A profile of epistemic states $\xi_i = (\mathcal{P}_i, r_i, \bar{\mu}_i, q_i, \sigma_i, k_i)$, $i \in N$—awareness, retained record, reference posterior, conjecture, committed policy, and failure counter, with acting beliefs μ_i derived by judged updating—is a* self-confirming unawareness equilibrium *(SCUE) if, under the play it induces,*

(i) each σ_i is subjectively optimal (equation (2));

(ii) each conjecture q_i agrees, in the limit of its empirical frequencies, with the play induced by σ_{-i} at every on-path cell;

(iii) no dissonance survives: each agent's active set, computed from its record as play accumulates, is almost surely eventually nonempty, and the acting-belief mass on any η-inadequate theory vanishes almost surely; and

(iv) no agent elects inquiry: $\lambda_i^{(k_i)} \widehat{B}_i(r_i) < \gamma_i$ along the induced records, which holds in particular whenever $\widehat{B}_i = 0$.

The record and failure counter belong in the state: dissonance status, leverage, and the achievability weight are functions of retained evidence and inquiry history, so two histories with identical partitions, beliefs, and policies but different records can differ in equilibrium status. Conditions (ii)–(iii) are stated asymptotically because empirical conjectures and stochastic tests fluctuate at finite samples. Deterministic consequence mechanisms make a contradictory test observation irreversible, but they do not eliminate sampling variation in states or in rivals' actions within an agent's awareness cell; empirical conjectures therefore generally still converge only asymptotically.

Conditions (i)–(ii) are subjective rationality with confirmed conjectures: the self-confirming conditions of the fixed-language tradition (Fudenberg and Levine, 1993; Kalai and Lehrer, 1993b). Conditions (iii)–(iv) are the new, epistemic ones: the language survives its own evidence, and the question has no priced value. A SCUE in which every partition is sufficient for τ and beliefs concentrate near the truth is simply a self-confirming equilibrium; the concept's interest lies entirely in the other cases.

Definition 9.2 (Dissonant stasis)**.** *Agent i is in* dissonant stasis *from some generation on if dissonance recurs while it never both elects and succeeds—because $\lambda_i^{(k_i)} \widehat{B}_i(r_i) < \gamma_i$ (no priced repair: zero leverage, or an abandoned question after the achievability weight has decayed, possibly at an irreparable record), or because elected production fails at every subsequent attempt. Its committed policy is carried forward with a standing anomaly. A population profile is* absorbing *if every agent is, from some generation on, either in a SCUE role (never dissonant) or in dissonant stasis.*

Remark 9.3 (The grain of truth, restored endogenously)**.** *Rational-learning convergence requires a grain of truth: the environment absolutely continuous with respect to beliefs (Kalai and Lehrer, 1993a). Under unawareness the condition can fail at the level of language—the truth may be inexpressible—and statistical dissonance is precisely the detector of that failure. Each successful refinement restores the grain* on the path of play*: post-restoration, some expressible theory matches the visited conditionals within tolerance. A SCUE is a configuration in which the grain of truth holds on-path for every agent and no one has a priced reason to look for the off-path remainder.*

Assumption 9.4 (Regularity)**.** *(i)* Generic tolerance. *In the stochastic case, the tolerance η lies outside the finite set of critical values at which, for some agent, theory, partition profile, and policy profile, the total-variation distance between an induced on-path conditional and the theory's prediction equals η. In the deterministic case the exact calibration $\eta = 0$, $\bar{n} = 1$, and $w \equiv 0$ applies instead, together with the deterministic conclusion of lemma 5.2. (ii)* Production randomness. *The randomization in each ϕ_i is independent across elections and of the state and consequence draws, and, conditional on an election at a* repairable *record ($\mathcal{R}(\mathcal{P}_i, \bar{r}_i) \neq \varnothing$), production succeeds with probability at least $\underline{\pi} > 0$. At an irreparable record production fails surely (definition 6.4).*

Part (i) excludes finitely many tolerance values in the stochastic case, since agents, theories, partitions, and policies are all finite, and handles the exact deterministic boundary separately. Part (ii) gives every elected repairable question a real chance of resolution; finiteness of elections comes instead from the decay of the achievability weight, so no separate never-produce exclusion is needed. The agent, unable to tell a repairable record from an irreparable one, may elect either; the two are distinguished only in their success probabilities, not in the agent's decision.

Theorem 9.5 (Epistemic absorption)**.** *Under assumption 9.4, almost surely:*

(a) at most $n(|S| - 1)$ refinements occur, and every agent's awareness partition is eventually constant;

(b) only finitely many inquiry elections occur—each agent stops electing once its achievability weight has decayed enough that $\lambda_i^{(k)}\, T(\max \pi_i - \min \pi_i) < \gamma_i$, and this happens after finitely many consecutive failures because $\widehat{B}_i$ is bounded and $\gamma_i > 0$; and

(c) there is an almost surely finite generation after which no agent's language changes and no agent pays an inquiry cost: every agent's epistemic life thereafter is judged-Bayesian updating within a fixed language, punctuated at most by dissonance episodes that fail the participation threshold.

Moreover, on the event that the committed profile is eventually constant,

(d) every agent's active set is eventually constant, equal to the theories η-adequate under the terminal play at every recurrent pair and satisfying every frozen constraint, and each agent is eventually either never dissonant—it lives in a Bayesian world—or in dissonant stasis (definition 9.2).

The proof is in appendix A.

Proposition 9.6 (Rest points and persistence)**.**

(i) Suppose awareness partitions and committed policies are eventually constant almost surely, conjectures and reference posteriors converge, and each non-stasis agent's terminal action is strictly optimal under its limiting belief and conjecture by a positive margin. Then the limiting configuration is a partial SCUE*: the SCUE conditions hold for every non-stasis agent, taking the stasis agents' constant carried-forward policies as fixed environment. (Belief coordinates need not literally freeze; the strict margin transfers optimality to the limit by continuity of the finite-dimensional payoff comparison.)*

(ii) Conversely, consider a configuration whose epistemic states satisfy the SCUE conditions, whose play generates only records with $\widehat{B}_i = 0$, and in which each agent's committed action at each cell is strictly optimal under every conjecture and every theory in its active set.

(a) With deterministic mechanisms, the configuration persists: dissonance, elections, and policy changes never occur, and the SCUE conditions continue to hold, with conjectures converging empirically to induced play.

(b) With stochastic mechanisms, suppose in addition that, at the entry configuration, every agent's active set is nonempty and every theory it contains is η'-adequate under the configuration's play for some $\eta' < \eta$. Suppose also that the strict optimality extends to every $(\eta + \varepsilon)$-adequate theory for some $\varepsilon \in (0, 2(\eta - \eta')]$. Then for every $\delta > 0$ there is a count threshold $N(\delta)$ such that the following holds. If the configuration is reached at a generation boundary and, at every recurrent pair (C, a),

$$n_i(C, a) \geq N(\delta) \qquad \textit{and} \qquad \mathrm{TV}(\hat{p}_i(\cdot \mid C, a), p_i^*(\cdot \mid C, a)) \leq \frac{\varepsilon}{4},$$

then, conditional on the entry history, with probability at least $1 - \delta$ play never changes, dissonance and elections never occur, and the SCUE conditions hold in every subsequent generation, with conjectures converging empirically to induced play.

The proof is in appendix A.

Remark 9.7 (What the theorem does and does not claim)**.** *Theorem 9.5 is a statement about the epistemic process—the death of inquiry—not about the convergence of play. Within a Bayesian world, committed policies may in principle continue to cycle as beliefs and conjectures chase one another; that is the classical convergence problem of learning in games (Fudenberg and Levine, 1993; Kalai and Lehrer, 1993a), it is orthogonal to this paper's contribution, and the theorem deliberately does not resolve it. Part (d) and the rest-point characterization are accordingly conditional on play stabilizing, which in every configuration featured in this paper is immediate: their beliefs make one action per cell strictly dominant. The decay of $\lambda_i^{(k)}$ is what makes inquiry die out even when the question is genuine: an agent facing a true, valuable, but hard-to-repair defect abandons it once repeated failure has eroded its hope. Abandoned* *questions are thus a substantive absorbing outcome, not an artifact—inquiry ceases because it is answered, because it is unpriced, or because it is given up. Were $\lambda_i^{(k)}$ held fixed instead of decaying, an agent could elect a doomed inquiry forever, paying γ_i each generation for a repair that never arrives; the decay rules this out and is the model's account of when a rational inquirer stops.*

Remark 9.8 (Rational anomaly tolerance)**.** *Zero-leverage dissonant stasis is a recognizable condition, not a pathology of the model: the agent registers that its causal account is broken—its test rejects every expressible theory—yet its deliberative valuation finds no coherent grouping with positive aggregate robust gain, so repairing the account is worth nothing to it. Standing anomalies tolerated by working scientists and managers have exactly this structure, and the model locates them: anomaly tolerance is rational precisely when the agent's coherent groupings carry no robust action leverage. Two sharper instances complete the taxonomy. At an irreparable record (remark 6.3) no expressible repair exists at any price—the anomaly is not merely unprofitable but impossible to fix within the current grammar—though the agent, unable to certify irreparability under noise, may keep electing while its deliberative record supports positive leverage and until the achievability weight decays. And an* abandoned question *is a repairable, positively-valued defect the agent stops pursuing because repeated failure has driven $\lambda_i^{(k)} \widehat{B}_i$ below cost: rational surrender, not indifference.*

Proposition 9.9 (On-path adequacy, and where error lives)**.** *In a SCUE, acting-belief mass on every η-inadequate theory vanishes almost surely. Under the genericity condition of lemma 5.2, or in its exact deterministic case, every theory remaining in the eventual support of μ_i is η-adequate at every on-path pair. All falsity retaining nonvanishing acting-belief mass is confined to what the configuration does not expose: mechanism assignments at states pooled with causally different states whose mixture remains within tolerance of a single mechanism at the visited profiles, and counterfactual claims at unvisited profiles.*

Proof. The first claim is definition 9.1(iii). Under genericity, or under the lemma's exact deterministic argument, lemma 5.2 makes the active set eventually constant and excludes every η-inadequate theory; judged updating restricts the acting belief to that set. The residual clause is the complement of the on-path constraints. □

Proposition 9.10 (Persistent performance differences)**.** *There exist primitives and absorbing configurations in which agents hold different terminal partitions and theories, expected per-generation payoffs differ strictly and permanently, and no disadvantaged agent has a priced repair: each such agent is in dissonant stasis, with $\lambda_i^{(k_i)} \widehat{B}_i(r_i) < \gamma_i$ from some generation on.*

The proof is by the instance below, and the mechanism deserves emphasis. Performance differences persist not because an equilibrium concept freezes them but because the process that would erase them has died economically: the disadvantaged agent's question is priced below the cost of asking—through zero leverage, an eroded achievability weight, or an irreparable record. Terminal heterogeneity in awareness is path dependence in the strong sense—agents with identical preferences and identical Bayesian competence, facing the same objective system, end at different languages and different payoffs because their histories posed different questions.

Definition 9.11 (Diagnostic environments)**.** *An expressible theory is* payoff-relevantly false *for agent i under a profile if the truth would render i's committed action suboptimal at some visited state—not merely shrink the set of optimal actions. The primitives* $(\Theta, \tau, \rho, \pi)$ *are* diagnostic *if, under every profile sustainable in a SCUE, every payoff-relevantly false theory is* η*-inadequate.*

Proposition 9.12 (No payoff-relevant error in diagnostic environments)**.** *In a diagnostic environment, every SCUE is free of payoff-relevant error. Payoff-relevant falsity survives only in environments that admit concealing profiles, and it requires them to be sustained: collective error needs the absence of* both *corrective forces, strategic variety inside agents' categories and environmental punishment of uniform misservice.*

Proof. By definition 9.1(iii), theories surviving in a SCUE are not η-inadequate; in a diagnostic environment every payoff-relevantly false theory is. The second claim restates definition 5.5: a false theory survives only under concealing profiles, and, under its no-legacy proviso, proposition 5.6 makes concealment necessary for survival and sufficient for avoiding eventual dissonance. □

Example 1 (continued). *The trap.* Set $\beta = 1$ and suppose both firms acquired the column refinement after discovery and became certain of the false column assignment. Each plays the column policy; the firms pool at every state; every payoff is one-half; and one-half is exactly what the column theory predicts under pooled play. The configuration is a SCUE with payoff-relevant error: policies are subjectively optimal, conjectures confirmed, and no dissonance ever occurs because the profile is concealing—so the technology never triggers, whatever its cost or reliability. The warrant gap of section 8 is permanent: unbounded experience accumulates while each firm remains exactly half-warranted, the missing half being the reflective weight on the world their shared language cannot express.

The knife edge. Now let $\beta < 1$. In the first generation containing an off-diagonal state, the pooled firms receive $\beta/2$ where their theory predicts one-half: the demand side reveals the label, dissonance returns, $\widehat{B}$ turns positive, and the reflective posterior becomes degenerate. Payoff-relevant self-confirmation is a knife-edge property of perfectly forgiving demand, and the destroyed value $1 - \beta$ at each misserved episode is itself the corrective signal: what demand withholds from the firms, their epistemics receives as evidence.

Persistence. Run the asymmetric configuration at $\beta < 1$: firm 1 holds the row refinement and the true assignment; firm 2 remains coarse and, by $\gamma_2 > \lambda_2^{(k_2)} \widehat{B}_2$, sits in dissonant stasis. Here firm 2's coarse record genuinely rejects all three constant theories—its differentiated losses defeat the constant theory matching its own action, its pooled payoff of one-half defeats the opposite theory's $\beta/2$ prediction, and neutrality fails at the differentiated losses—so it is dissonant rather than merely mistaken, and it stays put because its question is priced below cost. Firm 1 offers the preferred architecture in the row its rival misserves, winning both those states and splitting the rest, so under uniform ρ the firms earn three-quarters and one-quarter, permanently. (At $\beta = 1$ this instance instead sends firm 2 oscillating: whichever constant architecture it plays, the surviving constant theory prescribes the other, so its policy never settles—an instructive contrast, but not a persistent *stable* difference, which is why the diagnostic environment $\beta < 1$ is the right home for the result.) Every route out—strategic variety, demand harshness, a cheaper or more hopeful inquiry—is exactly one of the model's named forces.

10 Discussion

10.1 Implications for the theory-based view

The model supplies the theory-based view with the object its formal treatments presuppose and, in doing so, changes what several of the view's claims mean.

Strategic variety has a pre-probabilistic source. In the model, agents with identical preferences, identical Bayesian competence, and identical objective opportunities diverge because their experiences posed different questions, their technologies produced different repairs, and their positions extinguished inquiry at different points. Heterogeneous prior understanding generates heterogeneous questions; questions generate heterogeneous languages; languages generate heterogeneous theories, plans, and payoffs. This sequence begins before probability does, and it is path-dependent in the strong sense: the order of experiences shapes the terminal language, and agents end at different languages and different payoffs (proposition 9.10).

The value of experimentation is not information. An agent prices a question by whether some coherent reading of its own record supports a robustly improving action change (proposition 7.4), not by how much its resolution would reduce uncertainty. This reframes theory testing: high-entropy uncertainty with no action leverage is rationally ignored, and narrow contrasts that straddle an action threshold are rationally pursued. It also explains rational cessation—why successful firms stop asking—the loser's greater incentive to inquire, since competitive defeat concentrates the evidence and the dissonance in the defeated firm, and the abandonment of genuine but stubborn questions, as the achievability weight decays under repeated failure.

Early confidence outruns evidence even when correct. The warrant results discipline a claim at the center of the empirical program on scientific entrepreneurship (Camuffo et al., 2020; Zellweger and Zenger, 2023). A founder's theory that beautifully organizes the experiences that generated it has, by theorem 8.2, earned nothing by that organization: the fit was guaranteed by construction. Validation requires evidence whose likelihood differs across the candidate worlds—and, by corollary 8.3, the founder typically cannot even express the comparison, which is a formal argument for outside challenge (boards, investors, advisors with different awareness) as a warrant technology rather than a monitoring technology.

Markets discipline theories through two channels, and their absence is predictable. Payoff-relevant collective error requires both concealment by play and forgiveness by demand (proposition 9.12).

Industries where customers absorb misservice—captive buyers, no substitutes, third parties bearing the cost—can sustain shared false theories indefinitely; industries with strong outside options discipline theories even under consensus. Where demand is forgiving, the only corrective is strategic variety in beliefs, which gives belief heterogeneity a social value distinct from its private value: the entrant with a deviant theory is the industry's epistemic infrastructure.

Anomaly tolerance is an absorbing-state phenomenon. Firms that register their causal account is broken and rationally decline to fix it (remark 9.8) are not failing at learning; their standing anomalies carry no action leverage, or have grown too costly to keep pursuing. Such firms sit in dissonant stasis, outside any Bayesian world, which is why the condition is an absorbing state rather than an equilibrium. The model locates where tolerated anomalies accumulate: where consequences are opaque, so the agent cannot price the forgone value of a repair, and—even under transparency—near decisions whose robust optimum is insensitive to the unresolved structure.

10.2 The cognitional interpretation

The model's architecture tracks a classical account of inquiry (Lonergan, 1992; Ryall and Wilkins, 2018), and the correspondence is disciplining rather than decorative: it dictates which formal objects must not be conflated. Experience is the flow of episodes; the question for intelligence is dissonance, which localizes a defect without naming its repair; heuristic anticipation is the agent's search over coherent groupings of its own record, which gropes toward an answer without supplying its content; insight's formal footprint is the refinement (whose direction the analyst models through restoration, minimality, and the complexity ordering); formulation is the mechanism assignment on the new cells; reflective judgment is the agent's consistency test—the faculty distinct from the posterior that asks whether any expressible theory is adequate (remark 4.2), as against the analyst's warrant accounting, which the agent generally cannot perform on itself (corollary 8.3); and decision is the committed policy, which then determines what experience will teach next. The election adds the economic layer: an unrestricted desire to know, prosecuted under restricted resources, and eventually surrendered when hope of an answer fails. Nothing in the mathematics represents the conscious act of insight itself; the model represents its products and its price, which is what an economic theory can responsibly claim.

10.3 Limitations and the research program

Four boundaries of the present model define the program's next steps. First, the mechanism vocabulary Θ is fixed: agents lack distinctions, never concepts of consequence-generation. Expanding the vocabulary itself—deeper unawareness—is the natural second paper in the sequence, and the awareness-transition machinery here is built to receive it. Second, the model prices participation in inquiry but not its intensity or duration, and the technology's success is exogenous; endogenizing the depth of inquiry connects to models of costly deliberation and planning (Ryall, 2026), where cached prior conclusions govern when reconsideration is rational—the theory-to-plan transition. Third, agents are individuals; the theory-bearing *firm*—how insights become shared languages, collective intentions, and organizational commitments—is a separate paper resting on the same primitives (Ryall and Wilkins, 2018). Fourth, the competitive consequences here run through payoffs in a fixed game; joining the front end developed here to the cooperative back end of value capture (Gans and Ryall, 2017; Bryan et al., 2022), where awareness itself earns rents and disclosure is strategic, completes the arc. Within the present model, the flagged analytical items are general conditions for the convergence of play within Bayesian worlds (remark 9.7), equilibrium in the anti-coordination region that value destruction creates, counterfactual transparency as a design variable, and Markov state processes.

11 Conclusion

This paper models the step the theory-based view has described but not formalized: the origin of a strategic theory. The model draws a formal boundary between learning within a causal language and producing a new one, and it prices the crossing. Bayesian updating cannot expand the language and cannot even detect that expansion is needed; detection is statistical dissonance, a faculty distinct from the posterior. Dissonance directs repair without determining it; observational equivalence makes any economical inquiry technology fallible; and the agent estimates the value of a question—from its own record, honestly, without the analyst's list of answers—at exactly the moment the posterior is undefined. The return to Bayesian life is disciplined by a warrant account whose central theorem is easily stated: the insight is not evidence. And because theories cause the actions that cause the evidence, inquiry is almost surely absorbed: eventually every agent either lives in a Bayesian world or rests in a dissonance it has no priced reason to fix—in configurations that preserve false theories, heterogeneous languages, and permanent performance differences. Shared false theories survive exactly when neither strategic variety nor environmental harshness disturbs them; performance gaps persist whenever the laggard's question is priced below the cost of asking. Strategy's oldest question—why do similar firms perform differently?—here receives an answer that begins before beliefs: because they came to different questions, and some questions died, or were abandoned, before they were answered.

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A Proofs of Theorem 9.5 and Proposition 9.6

Probability framework. All randomness is carried by three mutually independent i.i.d. families on a common probability space: state draws $\{s_{g,t}\}$ over generations g and episodes $t \leq T$, with law ρ; consequence randomizers $\{U_{g,t}\}$, uniform on $[0,1]$, from which $x_{g,t}$ is generated through a fixed measurable representation of the kernel $\tau(s_{g,t})(a_{g,t})$; and production randomizers $\{V_{i,g}\}$, uniform on $[0,1]$ (assumption 9.4(ii)). Let $\mathcal{F}_g$ denote the σ-algebra generated by all randomness through generation g. Committed policies and election indicators for generation g are $\mathcal{F}_{g-1}$-measurable, since they are functions of records, beliefs, and conjectures formed from earlier generations. Conditional on $\mathcal{F}_{g-1}$ and on an election by agent i at a repairable record, production succeeds with probability at least $\underline{\pi}$; at an irreparable record it fails surely.

Part (a). Each successful production strictly refines one agent's partition, and, by cumulative awareness (assumption 7.1), no partition ever coarsens. The total cell count $\sum_i |\mathcal{P}_i|$ is integer valued, starts at no less than n, never exceeds $n|S|$, and strictly increases at every success. Hence at most $n(|S|-1)$ successes occur on every path, and each agent's partition is eventually constant.

Part (b). Fix agent i and let $\Delta\pi_i = \max \pi_i - \min \pi_i$. By proposition 7.4, $\widehat{B}_i \leq T\Delta\pi_i$. If $\Delta\pi_i = 0$, then $\widehat{B}_i = 0$ and no election occurs because $\gamma_i > 0$. Hence suppose $\Delta\pi_i > 0$. The election condition $\lambda_i^{(k)}\widehat{B}_i \geq \gamma_i$ fails whenever $\lambda_i^{(k)} < \gamma_i/(T\Delta\pi_i)$; since $\lambda_i^{(k)}$ is nonincreasing with $\lambda_i^{(k)} \to 0$ and $\gamma_i > 0$, there is a finite k_i^* beyond which no election occurs at failure count $k \geq k_i^*$. The failure counter resets only on a successful production, and by part (a) there are at most $n(|S|-1)$ successes across all agents. Between consecutive successes the counter climbs monotonically, so each agent elects at most k_i^* times per success-interval and hence finitely often in total; summing over agents, only finitely many elections occur on every path. (Assumption 9.4(ii) is not needed for election finiteness or for parts (c)–(d); it ensures that an elected repairable question has a positive conditional chance of resolution before abandonment.)

Part (c). Let G^* be the larger of the last refinement generation and the last election generation, almost surely finite by (a) and (b). After G^*, no language changes and no inquiry cost is paid. Because the election rule is mechanical—agent i elects exactly when it is dissonant and $\lambda_i^{(k_i)}\widehat{B}_i \geq \gamma_i$—every dissonance episode after G^* must fail the participation threshold. All other epistemic activity after G^* is judged-Bayesian updating within the fixed terminal language.

Part (d). The event $E = \{\text{committed profile eventually constant}\}$ is future-dependent, so we handle the conditioning by coupling rather than by conditioning on E directly. Decompose $E = \bigcup_{G_2,\sigma^*} E_{G_2,\sigma^*}$ over the countable set of (generation, deterministic profile) pairs, where E_{G_2,σ^*} is the event that the committed profile equals σ^* from generation G_2 on and $G_2 > G^*$. Fix (G_2, σ^*) and an agent i. Define an *auxiliary* process that plays σ^* in every generation from G_2 on, driven by the same i.i.d. draws $(s_{g,t}, U_{g,t})$; its observations are a function of those draws alone, so unconditional almost-sure statements about it carry no conditioning bias. On E_{G_2,σ^*} the actual and auxiliary observations coincide from G_2 on, so any almost-sure property of the auxiliary process holds almost surely on E_{G_2,σ^*}.

For the auxiliary process, call a pair (C, a) *recurrent* if $S_{\sigma^*}(C, a) \neq \varnothing$ and *frozen* otherwise. A frozen pair is visited only in the finite pre-G_2 prefix, so its count and constraint are constant. A recurrent pair is visited each episode with fixed probability $\rho(S_{\sigma^*}(C, a)) > 0$ independently, so its count diverges and, by the strong law on the finite space X_i, its cumulative empirical conditional converges to $p_i^*(\cdot \mid C, a)$; the finite prefix contributes a vanishing fraction. By assumption 9.4(i) no theory sits at distance exactly η, so each theory's constraint at each recurrent pair is eventually settled, and with finitely many theories and pairs the active set is eventually constant, equal to the theories satisfying every frozen constraint and lying within η at every recurrent pair—this is A_i^*, and its members are η-adequate under σ^* at every recurrent pair. If $A_i^* \neq \varnothing$, agent i is eventually never dissonant. If $A_i^* = \varnothing$, it is eventually always dissonant; since no election occurs after G^*, the mechanical rule gives $\lambda_i^{(k_i)}\widehat{B}_i < \gamma_i$ at every such generation, so agent i is in dissonant stasis (definition 9.2)—including the zero-leverage, abandoned-question, and irreparable cases. Taking the union over the countably many (G_2, σ^*) gives the claim on all of E. □

Proof of Proposition 9.6(i). Suppose awareness and committed policies are eventually constant almost surely, conjectures and reference posteriors converge, and each non-stasis agent's terminal action is strictly optimal under its limiting belief and conjecture by a positive margin, with the stasis agents' carried-forward policies fixed. Condition (i) of definition 9.1 holds at the limit: the committed policy solves equation (2) each generation, and by the strict margin and continuity of the finite-dimensional payoff comparison in (μ_i, q_i), the argmax is constant near the limit, so the eventually constant policy is optimal under the limiting belief and conjecture. For condition (ii), empirical conjectures converge to the frequencies induced by the eventually constant profile, giving limit agreement. For condition (iii), a non-stasis agent is eventually never dissonant by theorem 9.5(d), and judged updating keeps the acting belief supported on the eventual active set A_i^*, whose members are η-adequate at every recurrent pair, so the mass on any η-inadequate theory is eventually zero. For condition (iv), elections cease by theorem 9.5(b). The limiting configuration thus satisfies the SCUE conditions for every non-stasis agent with the stasis agents' play held fixed—a partial SCUE. □

Proof of Proposition 9.6(ii-a). With deterministic mechanisms, rejection is irreversible. Because the entry configuration satisfies definition 9.1(iii), its active set contains at least one 0-adequate theory: otherwise every currently active theory would eventually be rejected and, since refill is impossible, the active set would remain empty thereafter. A 0-adequate theory matches every on-path observation and is therefore never rejected. The active set consequently remains nonempty, although it may shrink as initially active inadequate theories are rejected.

Dissonance never occurs; $\widehat{B}_i = 0$ at every generated record by hypothesis, so no election or refinement occurs; and strict optimality under every conjecture and every theory in the entry active set fixes σ_i as that set shrinks and as reference posteriors and empirical conjectures evolve. Empirical conjectures converge to induced play, and any inadequate theory is eventually rejected. The SCUE conditions therefore continue to hold. □

Proof of Proposition 9.6(ii-b). Let g_0 be the entry generation and condition on the history $\mathcal{F}_{g_0}$. Couple the actual process to an auxiliary process that holds the configuration's committed profile fixed after g_0. At each recurrent pair, the auxiliary process's future consequences are i.i.d. with conditional law $p_i^*(\cdot \mid C, a)$.

Fix one recurrent pair, write n_0 for its entry count, and let $\hat{p}_{n_0+m}$ denote its cumulative empirical conditional after m additional visits. Because the consequence space is finite,

$$\mathrm{TV}(q, p) = \max_{D \subseteq X_i} |q(D) - p(D)|.$$

For a fixed $D \subseteq X_i$, let $S_m(D)$ be the sum of the first m post-entry centered indicators $\mathbf{1}\{x_i \in D\} - p_i^*(D \mid C, a)$. The entry-margin condition implies that

$$\sup_{m \geq 0} \mathrm{TV}(\hat{p}_{n_0+m}, p_i^*) > \frac{\varepsilon}{2}$$

can occur only if, for some D and some $m \geq 0$,

$$|S_m(D)| > \frac{\varepsilon n_0}{4} + \frac{\varepsilon m}{2}.$$

For either sign, $\exp\{\pm 4\varepsilon S_m(D) - 2\varepsilon^2 m\}$ is a nonnegative Hoeffding supermartingale. Crossing the displayed boundary makes this supermartingale at least $\exp\{n_0\varepsilon^2\}$. Ville's inequality, followed by a union bound over signs and subsets of X_i, therefore gives

$$\Pr\left(\sup_{m\geq 0} \mathrm{TV}(\hat{p}_{n_0+m}, p_i^*) > \frac{\varepsilon}{2} \,\middle|\, \mathcal{F}_{g_0}\right) \leq 2^{|X_i|+1} \exp\left(-n_0\varepsilon^2\right).$$

Let P be the number of recurrent pairs across agents and $d = \max_i |X_i|$. Choose $N(\delta)$ large enough that

$$w(n) \leq \frac{\varepsilon}{2} \quad \text{for all } n \geq N(\delta), \qquad \text{and} \qquad P\, 2^{d+1} \exp\left(-N(\delta)\varepsilon^2\right) \leq \delta.$$

A union bound then implies that, with conditional probability at least $1 - \delta$, every recurrent empirical conditional remains within $\varepsilon/2$ of its on-path conditional forever.

On this event, every theory active at entry continues to pass at every recurrent pair, because

$$\mathrm{TV}(\hat{p}_i, \text{prediction}) \leq \frac{\varepsilon}{2} + \eta' \leq \eta.$$

Its frozen constraints are unchanged, so every active set remains nonempty and dissonance never occurs. Conversely, every theory admitted at a later date is $(\eta + \varepsilon)$-adequate, since

$$\mathrm{TV}(\text{prediction}, p_i^*) \leq \eta + w(n) + \frac{\varepsilon}{2} \leq \eta + \varepsilon.$$

The extended strict-optimality hypothesis therefore fixes each committed action for every conjecture and every acting belief that judged updating can produce. Induction under the coupling shows that actual play coincides with the auxiliary fixed play. Hence no dissonance, election, or refinement occurs. Empirical conjectures converge to induced play, and lemma 5.2 implies that acting-belief mass on every η-inadequate theory vanishes. Thus the SCUE conditions continue to hold. $\square$

B Supporting computations for the running example

Dominance thresholds. At an awareness cell C, write $p^a = \mu_i(\{m : m(C) = \theta^a\})$ for $a \in \{0, 1\}$. Against a rival playing 0, the subjective expected-payoff difference between own actions 0 and 1 is $\frac{1}{2}p^0 - (1 - \frac{\beta}{2})p^1$; against a rival playing 1 it is $(1 - \frac{\beta}{2})p^0 - \frac{1}{2}p^1$; mixtures follow by linearity. Action 0 is strictly dominant on C if and only if $p^0 > (2 - \beta)p^1$, and symmetrically. Certainty in an architecture label makes the labeled action strictly dominant for every $\beta \in [0, 1]$, which is what sustains the committed policies in the trap configuration. At $\beta = 1$ the threshold is simple belief asymmetry, which is what differentiates the firms' opening policies.

Per-token gain and deliberative leverage. Consider a retained token whose coherent group pins mechanism θ^{a^*} while the carried-forward policy plays $a \neq a^*$ at the token's cell, at $\beta = 1$. Proposing a^*: if the rival plays a^*, the committed action yields 0 (differentiated loss) and a^* yields $\frac{1}{2}$ (pool); if the rival plays a, the committed action yields $\frac{1}{2}$ (pool) and a^* yields 1 (differentiated win). The per-token robust gain is $\frac{1}{2}$ in both cases—transparency makes $\Theta_i(g)$ a singleton, so the mechanism minimum is degenerate. A token whose group's optimal action equals the committed one contributes 0. Discovery record: two tokens, gains $\{\frac{1}{2}, 0\}$, average $\frac{1}{4}$, so $\widehat{B}_i = (T/|r_i|)\sum_t \Delta = (2/2)\frac{1}{2} = \frac{1}{2}$. After the validation generation of section 7 (serving the two off-diagonal states, $T = 2$), firm 1's four-token cumulative record has no group favoring a switch, so $\widehat{B}_1 = 0$; firm 2's four tokens split into two θ^0-groups and two θ^1-groups, of which two disagree with its column policy (gain $\frac{1}{2}$ each) and two agree (gain 0), average $\frac{1}{4}$, giving $\widehat{B}_2 = (2/4) \cdot 1 = \frac{1}{2}$. The durable content is the asymmetry $\widehat{B}_1 = 0 < \widehat{B}_2$, not the coincidence that $\widehat{B}_2$ equals the discovery value.

The trap and its knife edge. With both firms certain of the column assignment, the column policy pools at every state; every payoff is $\frac{1}{2}$; and under pooled play every mechanism consistent with the column cells predicts $\frac{1}{2}$ at $\beta = 1$. No token conflicts, active sets never empty, and $\widehat{B}_i = 0$; conditions (i)–(iv) of definition 9.1 hold. For $\beta < 1$, at an off-diagonal state the pooled action is objectively wrong and the realized payoff is $\beta/2$ against a prediction of $\frac{1}{2}$; the payoff $\beta/2$ uniquely reveals that the other architecture is preferred, while a payoff of $\frac{1}{2}$ would leave preferred and neutral both possible. The revealed conflict makes one column cell incoherent, the active set empties, and the reflective posterior over the row and column worlds becomes degenerate: dissonance, leverage, and warrant closure arrive together.

Persistence payoffs. Take $\beta < 1$, firm 1 playing the true row assignment, firm 2 coarse and playing a constant architecture c. The firms differentiate at the two states of the row that c misserves—firm 1 wins both—and pool at the other row's two states, giving firm 1 average payoff $\frac{3}{4}$ and firm 2 $\frac{1}{4}$ under uniform ρ. Firm 2's cumulative record rejects all three constant theories: its differentiated losses defeat the constant theory matching c, its pooled payoff of one-half defeats the opposite theory's $\beta/2$ prediction, and neutrality fails at the differentiated losses, so it is genuinely dissonant. Its deliberative leverage $\widehat{B}_2$ is positive but bounded, and stasis obtains whenever $\gamma_2 > \lambda_2^{(k_2)}\widehat{B}_2$. (At $\beta = 1$ the same constant-c record leaves the opposite constant theory adequate, which prescribes switching c; firm 2 then oscillates rather than resting, so the persistent stable difference is specific to the diagnostic environment $\beta < 1$.)